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Key Points:

- Basalt thicknesses in the Chang'E-5 landing region were estimated using penetrating craters, partially buried craters, and Gravity Recovery and Interior Laboratory (GRAIL) data
- The four basaltic layers have accumulated thicknesses of 9–175, 11–167, 19–478, and 0–800 m with an uptick in eruption rates at ~2 Ga
- The late-stage volcanism on the Moon could be sustained by enriched heat-producing elements or due to tidal heating or a thin crust layer

Supporting Information:

Supporting Information may be found in the online version of this article.

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Thicknesses of Mare Basalts in the Chang'E-5 Landing Region: Implications for the Late-Stage Volcanism on the Moon

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Abstract China's Chang'E-5 mission has collected lunar samples from a young mare plain with elevated thorium contents in the northeastern Oceanus Procellarum of the Moon. In this study, we first established a local stratigraphic column of the landing region based on the radial variation of the titanium content in the ejecta of large penetrating craters, and we found that the landing region was flooded by at least four layers of mare basalts. By using the latest imagery, spectral, topographic, and gravity data, we then estimated the basalt thickness of each layer based on the excavation depth of penetrating craters, the morphology of partially buried craters, and the effective density spectrum technique, which can help to better understand the late-stage volcanism on the Moon. Our results show that the four layers of basaltic lava flows in the landing region have accumulated thicknesses of 9–175, 11–167, 19–478, and 0–800 m. Combined with surface areas and ages of the basaltic units, we then calculated the eruption rate of mare basalts in the landing region, which shows an uptick at about 2 Ga. This result indicates a rejuvenated volcanic activity on the Moon, most likely due to the presence of heat-producing elements in the KREEP layer. Alternatively, the long-standing volcanism in this region could be also sustained by extra heating induced by the tidal force and/or a thin crust layer that favors the magma ascent.

Plain Language Summary For the first time in over four decades, lunar samples were brought back to the Earth by China's Chang'E-5 space program. The region visited by this mission is one of the youngest volcanic plains on the lunar surface and was flooded by at least four layers of lava flows. By using the excavation depth of impact craters, the shape of impact craters whose ejecta were buried by lava flows, and gravity observations, we later estimated the thickness of each basaltic layer. Our results show that the four layers of mare basalts in the landing region have accumulated thicknesses of 9–175, 11–167, 19–478, and 0–800 m. We then calculated the amount of lavas that erupted per unit time in the landing region and found an increase in the volcanic flux at about 2 billion years ago. This increased eruption rate shows that reactivated volcanism has occurred on the Moon, which may result from the heat released by the radioactive materials in the lunar crust. Alternatively, the tidal force exerted by the Earth and/or the presence of a thin crust may also contribute to the generation and ascent of magmas.

1. Introduction

In December 2020, China's Chang'E-5 spacecraft successfully touched down on the lunar surface in the north-eastern Oceanus Procellarum and then collected and brought back 1.731 kg lunar samples to the Earth (C. Li et al., 2021; Yang & Lin, 2021). Based on the spectral data acquired by the Kaguya/Multiband Imager (MI) and the Lunar Reconnaissance Orbiter/Wide Angle Camera (LROC/WAC) (Otake et al., 2012; Sato et al., 2017), the basaltic unit visited by the Chang'E-5 mission was mapped as the Em4 unit in the false color composite map (Qian, Xiao, Head, van der Bogert, et al., 2021). The regolith layer of the Em4 unit was found to be 1.5–8 m thick with a moderate TiO₂ content of 5–8 wt.% and a high FeO content of 17–18 wt.% (Jia et al., 2021; Qian, Xiao, Wang, et al., 2021) (Figure 1). Underlain by older Imbrian and Eratosthenian basaltic units, the Em4 unit was interpreted to have emplaced as high-flux sheet flows fed by Rima Sharp to the north and Rima Mairan to

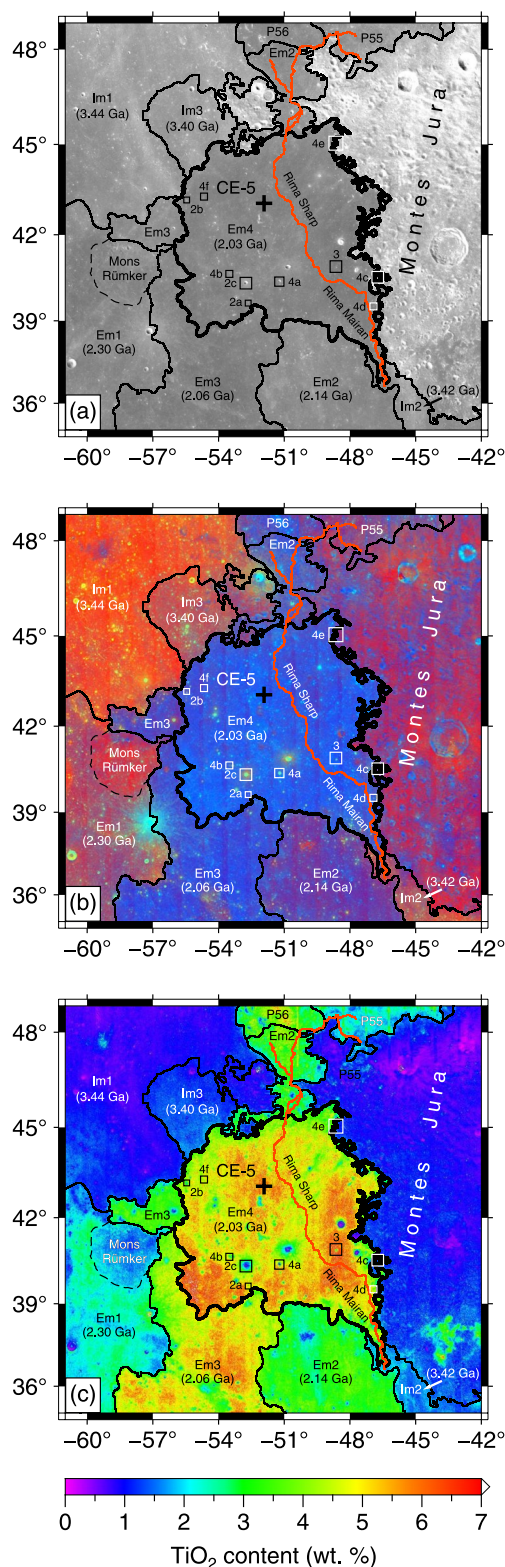


Figure 1. (a) Kaguya/TC mosaic, (b) Kaguya/MI false color composite, and (c) Kaguya/MI TiO₂ content map of the Chang'E-5 landing region and its surroundings. The Chang'E-5 landing site (black cross), Rima Sharp and Rima Mairan (red curve), Mons Rümker (dashed curve), and the boundaries of the Em4 (thick black curve) and other units (thin black curves) are denoted. Ages of the basaltic units were taken from Hiesinger et al. (2011), Q. Li et al. (2021), Wu et al. (2018), and this study. Locations of craters in Figures 2–4 are presented (boxes).

the South (Qian, Xiao, Head, & Wilson, 2021). Crater size-frequency distribution measurements of the Em4 unit indicated that this mare plain was formed between 1.2 and 2.2 Ga (e.g., Hiesinger et al., 2011; Jia et al., 2020; Qian et al., 2018; Qiao, Xu, Yang, et al., 2021). The latest radiometric dating of the returned Chang'E-5 basalts showed that the Em4 unit has an age of 1963 ± 57 Myr (Che et al., 2021) or 2030 ± 4 Myr (Q. Li et al., 2021). These results therefore confirmed that the Em4 unit hosts one of the youngest mare plains on the Moon, which could be ascribed to its location in the Procellarum KREEP Terrane (PKT) (Wieczorek & Phillips, 2000) that is enriched in the heat-producing elements (Qian, Xiao, Head, van der Bogert, et al., 2021). Previous observations from the Lunar Prospector/Gamma Ray Spectrometer (LP/GRS) have revealed elevated thorium contents of 5–8.5 ppm within the Em4 unit (Fu et al., 2021; Qian, Xiao, Head, van der Bogert, et al., 2021). Due to this increased amount of heat-producing elements, the melt production in the mantle can sustain for a few billions years (Laneuville et al., 2013, 2018; Wieczorek & Phillips, 2000). For this reason, the Em4 unit becomes a compelling target to study the late-stage volcanism on the Moon, which can help to better understand the thermal evolution of the lunar interior.

The thickness of lava flows, which can be used to constrain the volume of partial melts produced in the mantle, is key to understanding the thermal history of the Moon. The basalt thickness in the Chang'E-5 landing region was first investigated by using the excavation depth of craters that excavated underlying low-Ti basalts (Qian, Xiao, Head, van der Bogert, et al., 2021). In their study, the authors estimated the crater excavation depth as 0.084 times the crater diameter measured in the Kaguya/TC images and found that the summed basalt thickness of the Em3 and Em4 units is 39–63 m with a mean of 51 m. We noted that, however, the landing region was flooded by multiple layers of mare basalts (see Section 2.2) and that some of the large penetrating craters used by Qian, Xiao, Head, van der Bogert, et al. (2021) actually excavated much older Imbrian basalts at greater depths, suggesting that the basalt thickness of the Em3 and Em4 units was overestimated. Another issue that was not considered in Qian, Xiao, Head, van der Bogert, et al. (2021) is the way to estimate crater excavation depth from crater diameter. On the one hand, the crater excavation depth should be equal to 0.084 times the initial crater diameter (Melosh, 1989), whereas the present-day measured crater diameter is larger than its initial value as a result of crater degradation (Du et al., 2019). On the other hand, the apparent crater diameter measured in imagery data was found to be systematically smaller than the true rim-to-rim diameter measured in topographic data as the crater degrades (Turtle et al., 2005; Xie et al., 2017). A previous investigation of crater degradation in Mare Serenitatis shows that for a 2-Gyr-old crater with an initial rim-to-rim diameter of 500 m, its present-day rim-to-rim crater diameter would increase to 800 m and its present-day diameter measured in the optical image would be 600 m (Xie et al., 2017). It is still unclear how these two effects (overestimation of crater diameter when ignoring crater degradation vs. underestimation of crater diameter when using optical images) would affect the estimation of crater excavation depth in the Chang'E-5 landing region. Later on, Jia et al. (2021) also estimated the basalt thickness around the landing site based on the morphology of two nearby partially buried craters with diameters of 4.5 and 6.5 km. By calculating the difference between the rim height of a newly formed crater and the exposed rim height of the partially buried crater, the basalt thicknesses near the two craters were estimated to be 130 and 180 m. These results, however, are overestimates of the basalt thicknesses, as they used the rim height of a fresh crater to estimate the total crater rim height at the present day. The topographic degradation modeling of a 5-km diameter crater shows that its total rim height would reduce by over 30 m after 2 Gyr (Du et al., 2019).

In this study, we utilized penetrating craters, partially buried craters, and effective density spectrum to carry out a systematic estimation of the basalt thicknesses in the Chang'E-5 landing region. By synthesizing the results derived from these methods, we are able to obtain the thicknesses of different basaltic layers, which later allows us to delineate a more complete volcanic history of the landing region. The advances in our work compared with previous studies include the following: We (a) categorized penetrating craters into three classes, (b) took into account the crater topographic degradation, and (c) made use of the most up-to-date gravity models. We first introduced the spectral, imagery, topographic, and gravity data that were used in this study (Section 2.1). Then, the stratigraphic column of the landing region was established based on the ejecta composition of penetrating craters (Section 2.2). After this, we estimated the basalt thicknesses in the study region using the excavation depth of penetrating/nonpenetrating craters, the morphology of partially buried craters, and the effective density spectrum theory (Section 2.3). The estimated basalt thicknesses (Section 3) then enabled us to calculate the eruption rate of mare basalts onto the lunar surface, which gives insight into the late-stage volcanism on the Moon (Section 4).

2. Data and Methods

2.1. Spectral, Imagery, Topographic, and Gravity Data

Five orbital data sets were utilized in this study including spectral data acquired by the Kaguya/MI (Ohtake et al., 2008), optical images captured by the Kaguya/Terrain Camera (TC) (Haruyama et al., 2012), the digital terrain model (DTM) derived from the paired TC images via stereo-processing technique (Haruyama et al., 2012), the topographic model MoonTopo2600p obtained by the LRO/Lunar Orbiter Laser Altimeter (LOLA) (Wieczorek, 2007), and the gravity model GRGM1200B deduced from the Gravity Recovery and Interior Laboratory (GRAIL) observations (Goossens et al., 2020; Kahan, 2013). The Kaguya/MI obtained the surface reflectance of the Moon at nine visible and infrared wavelengths, and we used the algorithm in Lucey et al. (1998) and the parameters in Otake et al. (2012) to calculate the TiO_2 content of the study region. The resulting TiO_2 content map has a spatial resolution of 20 m/pixel and an uncertainty of 0.4 wt.% (Otake et al., 2012), which was used to identify penetrating and nonpenetrating craters larger than 100 m in diameter. The Kaguya/TC imaged the Em4 unit with a spatial resolution of 10 m (Haruyama et al., 2008), and the resulting images were utilized to identify partially and completely buried craters larger than 100 m in diameter. After stereo processing of the paired TC images, the generated Kaguya/TC DTMs cover the study area with a spatial resolution of 10 m and a vertical accuracy of 10 m (Haruyama et al., 2012). These stereo DTMs were utilized to extract the elevation profiles of penetrating/nonpenetrating craters and partially buried craters with diameters larger than 100 m, which were later used for topographic degradation modeling. The MoonTopo2600p topographic model is expressed in spherical harmonics of degree and order 2600 in a principal axis coordinate system and has a spatial resolution of 2.1 km (Wieczorek, 2007). The GRGM1200B gravity model is expanded to spherical harmonics of degree and order 1200 with an equivalent spatial resolution of 4.5 km (Goossens et al., 2020; Kahan, 2013). These topographic and gravity models were used to calculate the effective density spectrum.

2.2. The Stratigraphic Column of the Chang'E-5 Landing Region

As shown in the Kaguya/MI false color composite and TiO_2 content map (Figures 1b and 1c), the Em4 unit is surrounded by five volcanic units with different TiO_2 abundances. In this study, we named these units as Im1, Im2, Im3, Em2, and Em3 based on their progressively increasing TiO_2 contents. To retrieve the volcanic history of the Chang'E-5 landing region, it is necessary to construct its stratigraphic column that specifies the area, age, thickness, and volume of each basaltic unit.

First, the boundaries of these volcanic units were taken from Hiesinger et al. (2011) and Qian et al. (2018). We next determined their emplacement sequence by investigating the radial variation of the TiO_2 content in the ejecta of large penetrating craters. Here, we took crater Rümker H (40.33°N, 52.75°W; $D = 3.93$ km) in the Em4 unit as an example: From its crater rim to ejecta and to the background, the TiO_2 content shows a gradual increase from <1.2 wt.% (purple to deep blue) to 1.2–2.2 (cyan), 2.2–4.5 (green), and >4.5 wt.% (yellow to orange) (Figure 2c). According to the impact cratering theory, the closer the ejecta are emplaced to the crater rim, the deeper they were originally seated in the subsurface (Melosh, 1989). Therefore, it can then be inferred that older mare basalts have lower TiO_2 contents in the Chang'E-5 landing region. This correlation between older basalts and decreased TiO_2 contents was observed across the entire PKT, which can be explained by that, as the lunar interior cooled over time, the magma source moved from upper mantle down to core-mantle boundary where high-Ti materials accumulated after the mantle overturn (Kato et al., 2017; Qiao, Chen, Xu et al., 2021). In addition, it can be seen that the TiO_2 contents found in the ejecta of crater Rümker H are also found in basaltic units adjacent to Em4: Im1 has a TiO_2 content of <1.2 wt.%, Im2 and Im3 have TiO_2 contents of 1.2–2.2 wt.%, Em2 has a TiO_2 content of 2.2–4.5 wt.%, and Em3 has a TiO_2 content of >4.5 wt.% (Figure 1c). This apparent correlation certainly does not warrant that each radially zoned ejecta of crater Rümker H belong to a nearby basaltic unit with similar TiO_2 contents, as there could be some other unmapped basaltic units that are completely buried by Em4 and do not have any outcrops outside Em4. However, as a secondary consideration, it is still reasonable to assume that the ranges of TiO_2 contents found in the ejecta of crater Rümker H correspond to these surrounding basaltic units. Based on this assumption, it can then be inferred that the Im1, Im2, Im3, Em2, Em3, and Em4 units were emplaced in sequence. Then, we merged the Em3 and Em4 units and also combined the Im2 and Im3 units, as they share similar TiO_2 contents with each other (Figure 1c). Therefore, the six basaltic units investigated in this study were

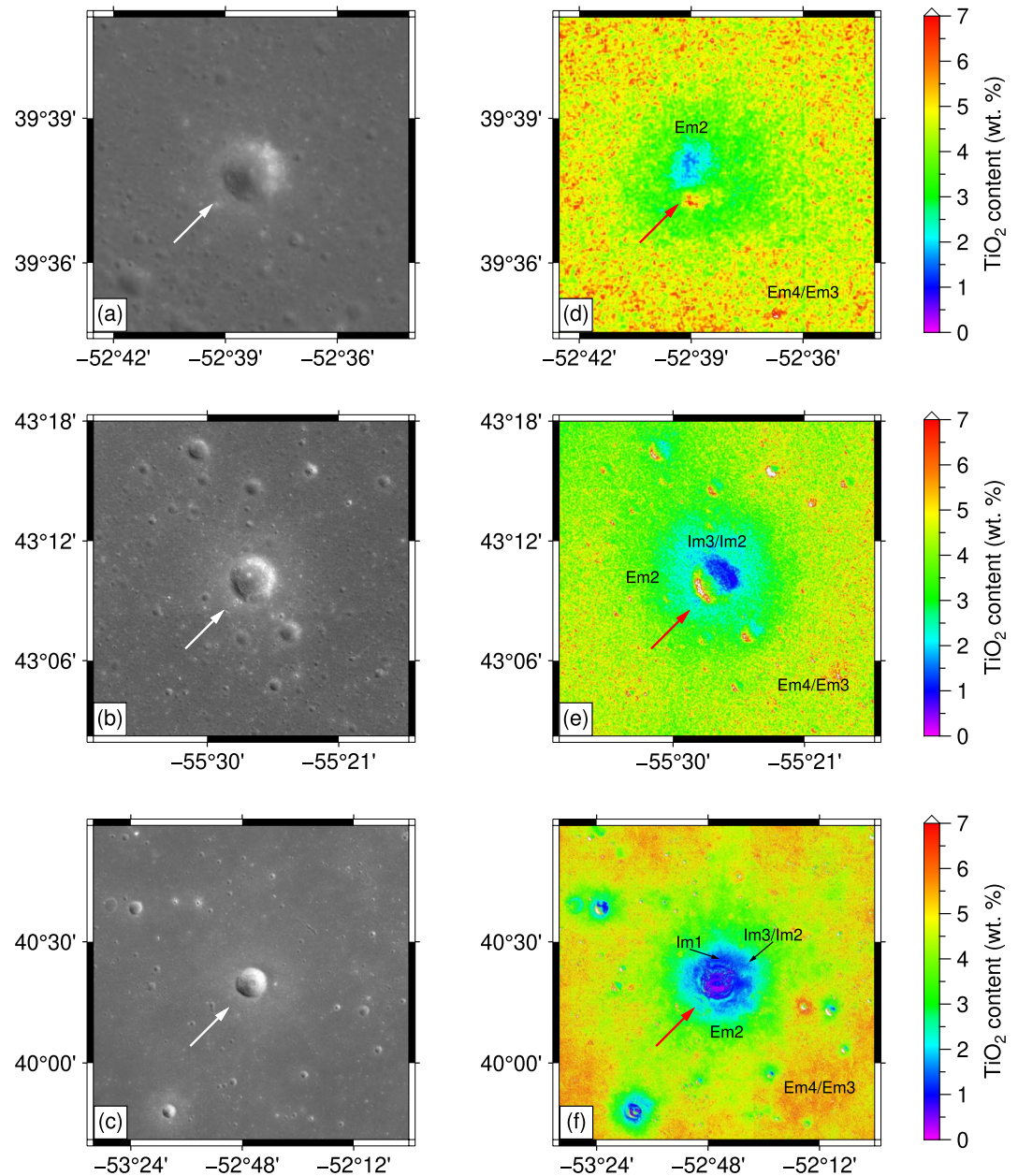


Figure 2. Kaguya/TC optical images and Kaguya/MI TiO₂ content maps of typical (a and d) Class A (39.63°N, 52.65°W; $D = 0.68$ km), (b and e) Class B (43.17°N, 55.45°W; $D = 1.14$ km), and (c and f) Class C (40.33°N, 52.75°W; $D = 3.93$ km) penetrating craters (white and red arrows) that excavated materials from Em4/Em3+Em2, Em4/Em3+Em2+Im3/Im2, and Em4/Em3+Em2+Im3/Im2+Im1 units. In (d–f), the Em4/Em3, Em2, Im3/Im2, and Im1 units are shown in yellow to orange, green, cyan, and purple to deep blue, respectively. Note that the color bars in (d–f) are the same as the one used in Figure 1c.

finally rearranged into four as Im1, Im2/Im3, Em2, and Em3/Em4 with TiO₂ contents of <1.2, 1.2–2.2, 2.2–4.5, and >4.5 wt.%, respectively.

Then, we assigned crater-counting ages in Hiesinger et al. (2011) and Wu et al. (2018) to the Im1, Im3, Em2, and Em3 units. For the undated unit Im2, we determined its surface age by counting craters in the Kaguya/TC mosaic and using production and chronology functions of Neukum et al. (2001) (Figures S1 and S2 in Supporting Information S1). The age of the Em4 unit was taken from the latest radiometric age of the returned Chang'E-5 basalts from Q. Li et al. (2021), as their result was derived from more measurements than Che et al. (2021). As a result, the Im1, Im2/Im3, Em2, and Em3/Em4 units were assigned with surface ages of 3.44, 3.42/3.40, 2.14, and

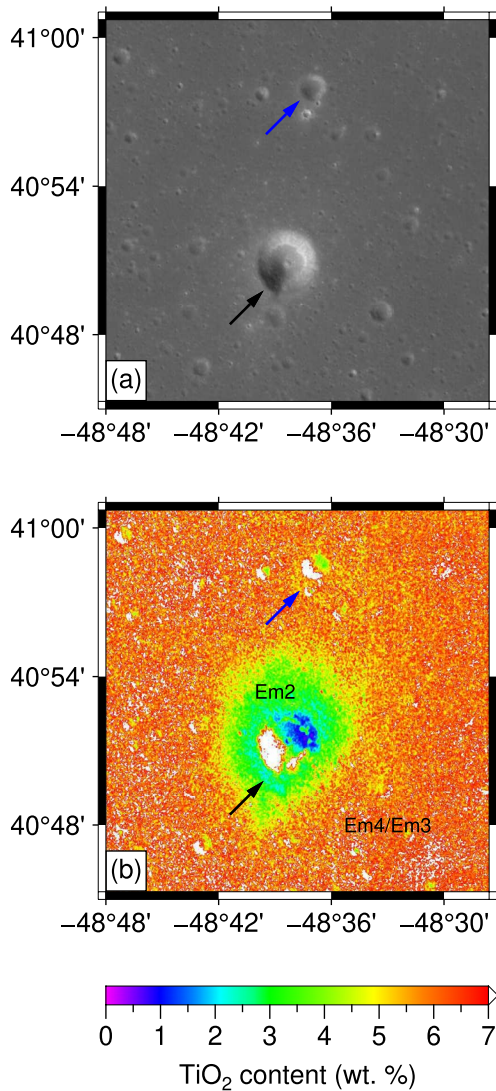


Figure 3. (a) Kaguya/TC image and (b) Kaguya/MI TiO_2 content map of paired penetrating (40.86°N , 48.64°W , $D = 1.24$ km; black arrow) and nonpenetrating (40.97°N , 48.62°W , $D = 0.54$ km; blue arrow) craters. Note that the color bar in (b) is the same as the one used in Figure 1c.

2.06/2.03 Gyr, respectively. It should be noted that there could be systematic biases in the six ages cited here as they were obtained by different authors and techniques. For this reason, the stratigraphic relations among different units were obtained based on the ejecta compositions of large craters instead of the absolute age of each unit. These absolute ages of the basaltic units are used later when calculating the eruption rates of mare basalts.

2.3. Basalt Thickness Estimation Methods

In this study, we use three different methods to estimate the basalt thicknesses in the Chang'E-5 landing region. First, the crater excavation depth technique is utilized to constrain the thickness of basaltic layers that craters have penetrated through. Then, the morphology of partially buried craters is applied to estimate the thickness of basalts that were emplaced after the crater formation. Lastly, the effective density spectrum is used to invert the total basalt thickness based on the density contrast between mare basalts and the highland substrate. As a result, we are able to estimate the thickness of basaltic layers at different depths, which can be used to construct a more complete stratigraphic column of the Chang'E-5 landing region.

2.3.1. Excavation Depth of Penetrating Craters

Using the Kaguya/MI TiO_2 content map (Figure 1c), the penetrating craters were identified if they have penetrated through the uppermost Em4/Em3 unit and excavated low-Ti materials from the subsurface. In order to estimate the thicknesses of mare basalts at different depths, each identified penetrating crater was then assigned to one of the following three classes. Craters of Class A excavated the Em4/Em3 and Em2 units as evidenced by their proximal ejecta with a TiO_2 content of 2.2–4.5 wt.%, whereas their distal ejecta display a TiO_2 abundance of >4.5 wt.% (Figure 2a). Class B represents craters that penetrated into the Em4/Em3, Em2, and Im3/Im2 units, and the TiO_2 content of their crater ejecta shows a radial transition from 1.2–2.2 to 2.2–4.5 and to >4.5 wt.% (Figure 2b). Class C craters dug out materials from units Em4/Em3, Em2, Im3/Im2, and Im1, and their ejecta exhibit TiO_2 contents of <1.2, 1.2–2.2, 2.2–4.5, and >4.5 wt.% from the crater rim to the background (Figure 2c). For each identified penetrating crater, we then looked for its corresponding nonpenetrating crater. In this study, the nonpenetrating crater was identified as the largest crater (blue arrow) that did not excavate any low-Ti materials in the 5 km diameter vicinity of the penetrating crater (black arrow) (Figure 3).

For each pair of penetrating and nonpenetrating craters identified in the TiO_2 content map, we next remapped them in the corresponding TC images that have a better spatial resolution and detailed textures in favor of crater identification. This was done by selecting three points that are 120° apart along the crater rim, and then the diameters of penetrating and nonpenetrating craters were fitted automatically using the Crater Helper Tools (Nava, 2011). At this point, the crater excavation depth was temporarily estimated as one tenth of the transient crater diameter that is 0.84 times the crater diameter measured in the TC image. Then, the upper ($T_{\max}$) and lower ($T_{\min}$) bounds of the lava flow thickness were obtained from the excavation depths of the paired penetrating and nonpenetrating craters:

$$T_{\max} = 0.1 \times 0.84 \times D_p \quad (1a)$$

$$T_{\min} = 0.1 \times 0.84 \times D_{np} \quad (1b)$$

where D_p and D_{np} are the diameters of penetrating and nonpenetrating craters.

In this manner, the accumulated thickness of the basaltic unit (Em4/Em3, Em4/Em3+Em2, and Em4/Em3+Em2+Im3/Im2) was constrained by the excavation depths of the smallest nonpenetrating and the largest penetrating craters (Class A, B, and C) found within the unit, respectively.

Next, we investigated if the above crater excavation depth, which was derived from the present-day crater diameter measured in the optical image, needs to be better constrained. First, we looked into the overestimation of the initial crater diameter that is induced by crater topographic degradation (Text S1 in Supporting Information S1) (Du et al., 2019). In this study, the topographic evolution of the crater was modeled by a diffusion equation that is characterized by an initial crater diameter and a degradation state. By solving the topographic diffusion equation, a look-up table was then set up that correlates the input parameters with a modeled crater elevation profile. The best estimate of the initial crater diameter can therefore be obtained by looking for the modeled profile in the look-up table that has the smallest root mean square (RMS) misfit with the observed profile. Second, we quantified the underestimation of the crater diameter in the optical image. We remapped all identified penetrating and nonpenetrating craters from TC images to TC DTM and then produced an azimuthally averaged elevation profile for each crater. The rim-to-rim diameter was later defined as twice the radial position of the rim crest in the elevation profile. Taken together, in the end we were able to compare the present-day crater diameter measured in the optical image, the initial rim-to-rim crater diameter, and the present-day rim-to-rim crater diameter, among which the initial rim-to-rim crater diameter is the one that should be used to calculate the crater excavation depth (Melosh, 1989).

2.3.2. Morphology of Partially Buried Craters

Based on how much of the crater rim is still visible, the crater whose distal ejecta were buried by lava flows can be divided into three classes: rim completely exposed (Figures 4a–4c), rim partially exposed (Figures 4d and 4e), and rim completely buried (Figure 4f) (Du et al., 2019). We identified the first two classes of craters within the Em4 unit if (a) the extent of their exposed ejecta is less than that for an unburied crater (about one crater radius for lunar craters) (Moore et al., 1974) and (b) the exposed ejecta have a distinct boundary with the background lava flow (white arrows in Figure 4). We also mapped the third class of craters (i.e., ghost craters) by a subtle topographic depression with a featureless buried rim that has a similar reflectance to the surrounding lava flow.

In this study, we made use of partially buried craters whose rims are completely exposed to estimate the basalt thickness (Text S2 in Supporting Information S1) (Du et al., 2019). This is because we assumed an axisymmetrical topography for the crater and its vicinity in the degradation model and the topographic diffusion equation was solved in a cylindrical coordinate. We then modeled the topographic degradation of these craters before and after the lava flow emplacement, and we made a simple assumption that multiple layers of lava flows (if any) were emplaced at the same time. At the beginning, the crater formed on the lunar surface with an initial crater diameter (D_{init}), which is the initial condition of the degradation model. This crater was later eroded to a certain degradation state (K_{pre}) just before the lava flooding and then the crater ejecta were suddenly flooded by lava flows with a given thickness (T). Next, the flooded crater elevation profile became the new initial condition, and as time went on, the crater continued to degrade to its present-day degradation state (K_{tot}). By varying D_{init} , K_{pre} , K_{tot} , and T with certain steps and ranges (Table S3 in Supporting Information S1), we next searched for the parameter combination that minimized the RMS misfit between the modeled (H_{mod}) and observed (H_{obs}) crater elevation profiles:

$$\sigma = \sqrt{\sum_{i=1}^N [H_{\text{mod}}(D_{\text{init}}, K_{\text{pre}}, K_{\text{tot}}, T; i) - H_{\text{obs}}(i)]^2 / N} \quad (2)$$

where i is the i th of N points along the profile.

The best estimate of the lava flow thickness was determined by the modeled profile with the smallest misfit. Then, the uncertainty of the basalt thickness was determined by the thickness estimates whose corresponding RMS misfits are smaller than the standard deviation of the observed elevation profile (see Text S2 in Supporting Information S1).

2.3.3. Effective Density Spectrum

The detected lunar free-air gravity is strongly dependent on the density profile of the subsurface, and larger gravity anomalies are usually found to be correlated with the presence of denser materials (e.g., Andrews-Hanna

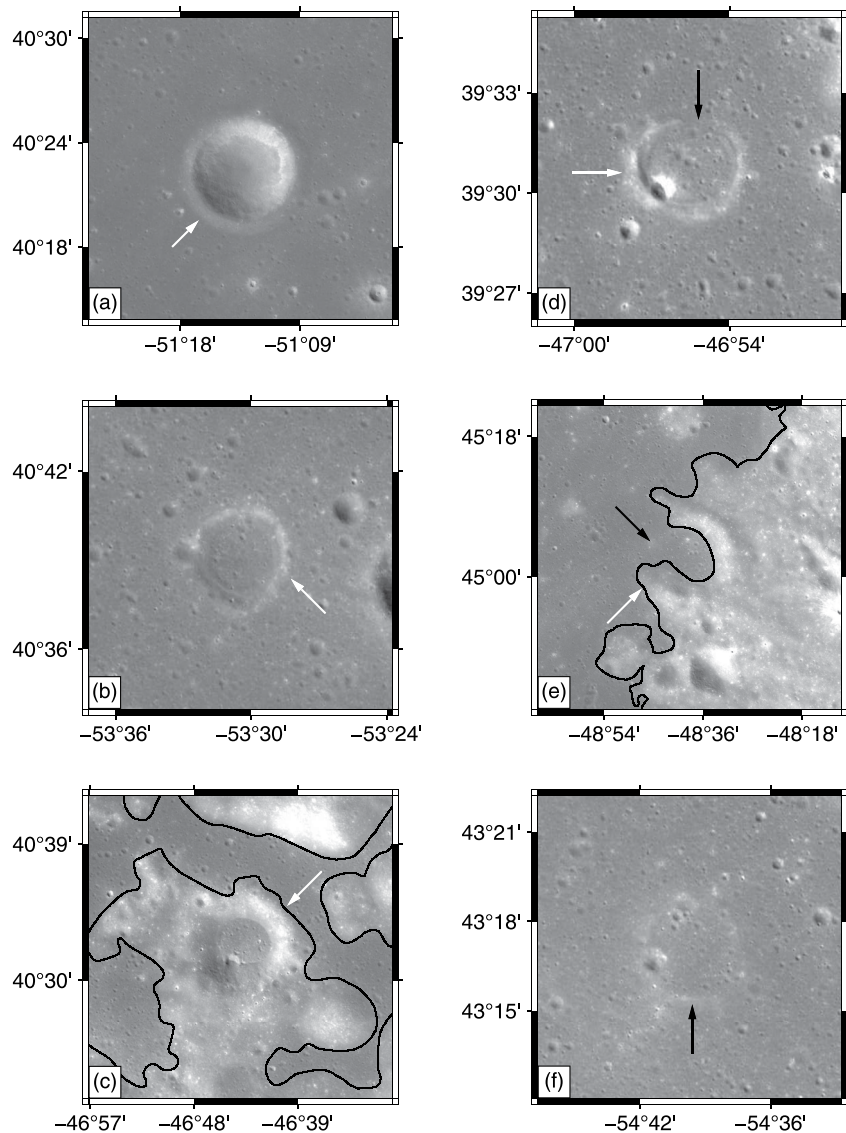


Figure 4. Kaguya/TC images of buried craters with rim completely exposed (a: 40.38°N , 51.22°W , $D = 3.19$ km; b: 40.65°N , 53.51°W , $D = 1.94$ km; and c: 40.54°N , 46.73°W , $D = 3.73$ km), partially exposed (d: 39.51°N , 46.93°W , $D = 1.53$ km and e: 45.04°N , 48.64°W , $D = 5.63$ km), and completely buried (f: 43.29°N , 54.66°W , $D = 1.72$ km). Craters in (a, b, d, and f) are located on mare whereas those in (c and e) are located along the mare-highland boundary (black curves). The boundary between the exposed crater rim and the lava flow is given by the white arrow and the buried crater rim is marked by the black arrow.

et al., 2013; Zuber et al., 2013). It was previously reported that lunar mare basalts ($3,010\text{--}3,270\text{ kg/m}^3$) and the underlying anorthositic highland substrate ($2,200\text{--}2,600\text{ kg/m}^3$) have distinct bulk densities (e.g., Kiefer et al., 2012; Wiczorek et al., 2013), and for this reason, it is possible to invert the thickness of the entire mare basalts based on their gravity signatures. In this study, the effective density spectrum derived from the GRAIL and LOLA data (Figures 5a and 5b) is introduced to estimate the total basalt thickness in the Chang'E-5 landing region. First, the modeled effective density spectrum can be obtained by integrating the density profile of the lunar subsurface, which includes an overlying basaltic unit and an underlying highland substrate. Second, the observed effective density spectrum is estimated as the ratio of the cross-power spectrum of observed free-air gravity and topography-related gravity to the auto-power spectrum of topography-related gravity. These calculated power spectra are later sampled by multiple tapers as the study area is only a small portion of the lunar surface. In the end, the best estimate of total basalt thickness can be found by comparing the modeled and observed effective density spectra (see Text S3 and Figure S10 in Supporting Information S1).

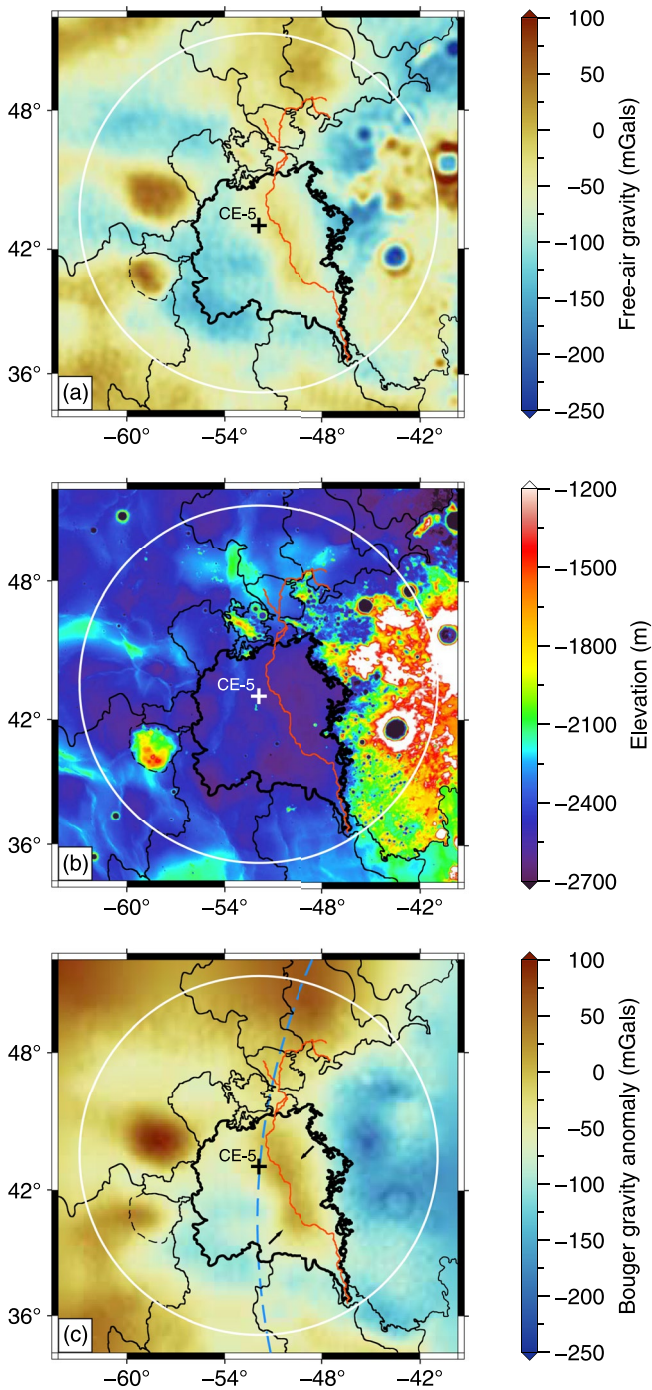


Figure 5. (a) Gravity Recovery and Interior Laboratory (GRAIL) free-air gravity, (b) LRO/Lunar Orbiter Laser Altimeter (LOLA) topography, and (c) GRAIL Bouguer gravity anomaly maps of the Chang'E-5 landing region. The white circles denote the 8° localization taper used in the effective density spectrum analyses. The red curves show Rima Sharp and Rima Mairan. In (c), the fourth ring of the Imbrium Basin ($D_{\text{ring}}^{\text{4th}} = 1,700$ km, centered at 35°N, 17°W) is delineated by the dashed blue curve (Spudis et al., 1988), and the region with a positive gravity anomaly is shown by black arrows.

In this study, the subsurface beneath the Chang'E-5 landing region was assumed to be composed of two layers, which include an overlying basaltic unit with a constant bulk density and an underlying highland substrate with bulk densities increasing with depth. As a result, the modeled effective density spectrum ($\rho_{\text{eff}}^{\text{mod}}$) can be obtained by integrating the density profile with depth, which is expressed as a function of total thickness of mare basalts (T_{total}), density of the first layer in the highland substrate (ρ_{sub}^1), density of mare basalts (ρ_{basalt}), and density gradient in the highland substrate ($d\rho_{\text{sub}}/dz$) (Besserer et al., 2014):

$$\rho_{\text{eff}}^{\text{mod}}(l) = \rho_{\text{basalt}} + (\rho_{\text{sub}}^1 - \rho_{\text{basalt}}) \left(\frac{R_0 - T_{\text{total}}}{R_0} \right)^{l+2} + \sum_{i=2}^N (d\rho_{\text{sub}}/dz) (r_i - r_{i-1}) \left(\frac{r_i}{R_0} \right)^{l+2} \quad (3)$$

where l is the spherical harmonic degree, N is the number of layers in the highland substrate, r_i is the spherical radius between the center of the Moon and the upper interface of the i th layer in the highland substrate, and R_0 is the average radius of the Moon within the localized taper (see below).

For the spherical harmonic of degree l and order m , the observed lunar free-air gravity (g_{lm}^{obs}) can be decomposed into two components with one (g_{lm}^{topo}) that can be predicted by unit-density topography and the other one ($g_{lm}^{\text{non-topo}}$) that is not related to topography (Besserer et al., 2014; Gong et al., 2016; Wiczeorek et al., 2013):

$$g_{lm}^{\text{obs}} = \rho_{\text{eff}}^{\text{obs}}(l) \cdot g_{lm}^{\text{topo}} + g_{lm}^{\text{non-topo}} \quad (4)$$

where $\rho_{\text{eff}}^{\text{obs}}$ is the observed effective density spectrum.

Note that the observed gravity coefficient was provided as a GRAIL data product (Goossens et al., 2020; Kahan, 2013), and the predicted gravity coefficient was calculated based on the MoonTopo2060p topographic model (Wiczeorek, 2007). We then multiplied Equation 4 by g_{lm}^{topo} , and since g_{lm}^{topo} and $g_{lm}^{\text{non-topo}}$ are not correlated, the second term on the right-hand side of the Equation 4 disappeared. As a result, an unbiased estimate of the observed effective density spectrum was written as the cross-power spectrum between the observed and predicted gravity coefficients divided by the auto-power spectrum of the predicted gravity coefficient ($\rho_{\text{eff}}^{\text{obs}} = S_{\text{global}}^{\text{obs-topo}} / S_{\text{global}}^{\text{topo-topo}}$).

As we aimed to study the geophysical response of a small region on the lunar surface, a localized multitaper analysis was conducted by using the SHTools (Wiczeorek & Meschede, 2018). As a result, the two calculated global power spectra were sampled ($S_{\text{win}}^{\text{obs-topo}}$ and $S_{\text{win}}^{\text{topo-topo}}$) by a taper with a given angular radius and bandwidth. The angular radius of the taper was determined by the study area and the bandwidth of the taper was carefully weighed after several trade-off trials: A relatively small bandwidth would result in too few well-localized tapers whereas a comparatively large bandwidth would neglect detailed variations of the signals. Furthermore, to prevent the loss of spectral power, each selected taper should concentrate at least 99% of its spectral power within the angular radius (Figure S11 in Supporting Information S1). Then, a given number of tapers were chosen for sampling and the final observed effective density spectrum is rewritten as the average of all individual localized spectral estimates:

$$\rho_{\text{eff}}^{\text{obs}}(l) = \frac{1}{K} \sum_{k=1}^K \frac{S_{\text{win},k}^{\text{obs-topo}}(l)}{S_{\text{win},k}^{\text{topo-topo}}(l)} \quad (5)$$

where k represents the k th of K tapers.

Lastly, the reduced chi-square function (χ_{red}^2) was used to evaluate the difference between the modeled and observed effective density spectra:

$$\chi_{\text{red}}^2 = \frac{1}{\nu} \sum_{l=l_{\min}}^{l_{\max}} \left[\frac{\rho_{\text{eff}}^{\text{mod}}(T_{\text{total}}, \rho_{\text{sub}}^l, \rho_{\text{basalt}}, d\rho_{\text{sub}}/dz; l) - \rho_{\text{eff}}^{\text{obs}}(l)}{\sigma^{\text{obs}}(l)} \right]^2 \quad (6)$$

where σ^{obs} is the standard deviation of the individual effective densities, $\nu = l_{\max} - l_{\min} - 2$ is the number of degrees of freedom, and $l_{\max}$ and $l_{\min}$ are the maximum and minimum spherical harmonic degrees determined by the correlation between gravity and topography.

The minimum misfit yielded the best fitting modeled effective density spectrum and hence the best estimate of the total basalt thickness. Then, a series of synthetic effective density spectra were generated based on the statistics of the gravity component that is not related to the topography, and the uncertainty of the obtained total basalt thickness was evaluated based on the cumulative distribution of the misfits between the best fitting and synthetic effective density spectra (see Text S3 in Supporting Information S1).

3. Thicknesses of Mare Basalts in the Chang'E-5 Landing Region

3.1. Excavation Depth of Penetrating Craters

In total, we identified 285 penetrating craters and 208 nonpenetrating craters within the Em4 unit. Their spatial distributions (Figures 6a–6c) show that most of them are located along the western and southern boundaries of the Em4 unit, where basalts are thinner and therefore penetrating craters are easier to form. The scarcity of penetrating craters on the northern and eastern margins of the Em4 unit is likely due to the contamination of low-Ti materials from the nearby highlands (Figure 1c), making penetrating craters (also with a low titanium abundance in the ejecta) more difficult to identify. Overall, the diameters of penetrating and nonpenetrating craters measured in TC images range from 0.14 to 5.69 km and from 0.10 to 0.77 km with medians of 0.56 and 0.30 km, respectively (Figures 6d–6f and Table 1). Among them, 234 craters (Class A) that dug out materials of the Em4/Em3 and Em2 units have diameters of 0.14–2.09 km with a median of 0.54 km, and the corresponding nonpenetrating craters have diameters of 0.10–0.77 km with a median of 0.29 km (Figure 6d). For 46 craters (Class B) that excavated the Em4/Em3, Em2, and Im3/Im2 materials, they have crater diameters of 0.20–1.99 km with a median of 0.68 km, and their paired nonpenetrating craters have diameters of 0.13–0.76 km with a median of 0.29 km (Figure 6e). For the five craters (Class C) that penetrated through the Em4/Em3, Em2, Im3/Im2, and Im1 layers, their crater diameters span a range of 1.66–5.69 km with a median of 2.31 km, and the associated nonpenetrating craters have diameters of 0.22–0.69 km with a median of 0.51 km (Figure 6f). We did not find any craters that penetrated through the entire basaltic layers and excavated underlying anorthositic materials from the highland substrate.

The upper and lower bounds of the lava flow thickness within a basaltic unit were given by the excavation depths of the smallest nonpenetrating crater and the largest penetrating crater. As a result, the summed basalt thicknesses of the Em4/Em3, Em4/Em3+Em2, and Em4/Em3+Em2+Im3/Im2 units were estimated to be 9–175, 11–167, and 19–478 m, respectively. While these thickness ranges are the primary, unbiased estimates for the basalt thickness, we also derived more straightforward basalt thickness estimates based on their statistics, which can later help us to make further geological interpretations. This was done by simply taking the average of the upper and lower bounds of the basalt thickness as the basalt thickness at the location of the paired penetrating/nonpenetrating craters (Figure 7) and then using the median value of the thickness estimates within the basaltic unit to represent the basalt thickness of this unit. As a result, the median basalt thicknesses of the Em4/Em3, Em4/Em3+Em2, and Em4/Em3+Em2+Im3/Im2 units were calculated to be 36_{-17}^{+17} , 40_{-23}^{+23} , and 110_{-80}^{+80} m, where the uncertainties are the standard deviations of the basalt thicknesses within the unit. Then, we can separate the thickness of Em2 from the combined thickness of Em4/Em3+Em2 and can subtract the thickness of Im3/Im2 from the summed thickness of Em4/Em3+Em2+Im3/Im2. As a result, the median basalt thicknesses of the Em4/Em3, Em2, and Im3/Im2 units were estimated to be 36_{-17}^{+17} , 4_{-4}^{+16} , and 70_{-70}^{+76} m, where the uncertainties were derived based on the error propagation law assuming a Gaussian probability distribution for the basalt thickness. Note that our estimated basalt thickness for the Em4/Em3 layer is on the lower end of 39–61 m in Qian, Xiao, Head, van der Bogert, et al. (2021). This

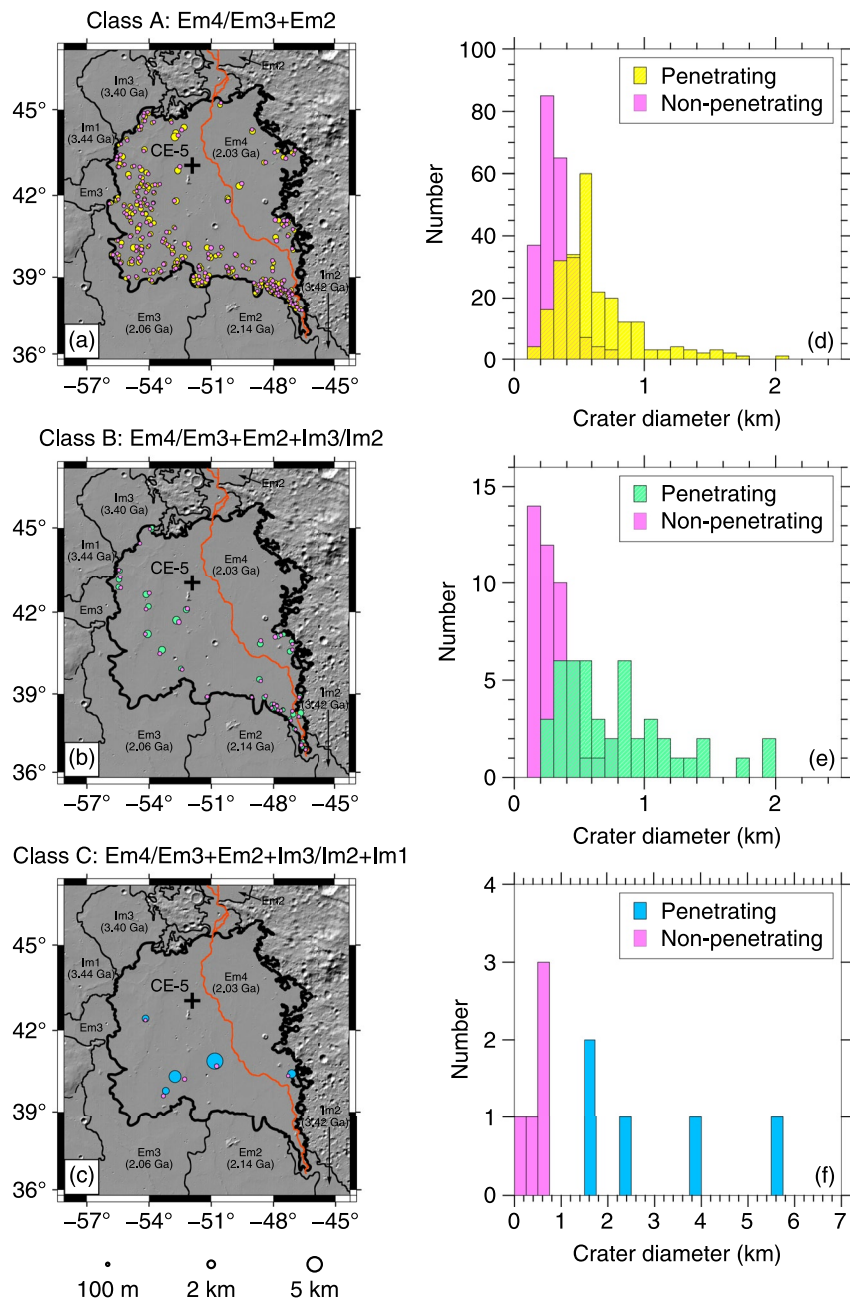


Figure 6. Spatial distributions of (a) Class A (yellow), (b) Class B (green), and (c) Class C (blue) penetrating craters and their corresponding nonpenetrating craters (pink). The crater diameters are scaled for visibility. Diameter histograms of (d) Class A (yellow), (e) Class B (green), and (f) Class C (blue) penetrating craters and their corresponding nonpenetrating craters (pink).

is because Qian, Xiao, Head, van der Bogert, et al. (2021) did not exclude the large penetrating craters that also excavated the Im3/Im2 and Im1 units when estimating the thickness of the Em4/Em3 unit.

We next quantified and compared the overestimation of initial crater diameter when ignoring the crater degradation and the underestimation of crater diameter when using optical images. It was then found that the topographic profiles of some craters were not well fitted by the topographic degradation model, most likely as a result of their unique geological contexts for crater formation and topographic degradation. Therefore, we chose 159 penetrating and 75 nonpenetrating craters whose RMS misfits are smaller than 10 m for further analyses (Figure S4 in Supporting Information S1) (Du et al., 2022). For the selected 234 penetrating and nonpenetrating craters (Table

Table 1
Classification of Identified Penetrating and Nonpenetrating Craters

Class	A		B		C	
Number of pairs	234		46		5	
Crater type	Penetrating	Nonpenetrating	Penetrating	Nonpenetrating	Penetrating	Nonpenetrating
Diameter range (km)	0.14–2.09	0.10–0.77	0.20–1.99	0.13–0.76	1.66–5.69	0.22–0.69
Median diameter (km)	0.54	0.29	0.68	0.29	2.31	0.51

S1 in Supporting Information S1), our inversion showed that their best fitting initial crater diameters have a range of 0.17–4.00 km with a median of 0.54 km. In the TC images, the measured crater diameters range from 0.13 to 3.93 with a median of 0.52 km. As a comparison, their present-day crater diameters defined by the elevation profiles span from 0.18 to 4.01 km with a median of 0.65 km.

The above results show that due to the crater topographic degradation process, the initial crater diameter is $\sim 17\%$ smaller than the present-day crater diameter defined by the elevation profile. Meanwhile, it was also found that the crater diameter measured in the optical image is $\sim 20\%$ smaller than that defined by the elevation profile. This discrepancy between crater diameters determined from imagery and elevation data can be explained by the fact that the rim crest of a degraded crater in the optical image tends to move into the crater interior in human eyes. This is because, for a degraded crater, the crater inner wall has larger variations of surface slopes hence a more significant transition between light and shadow in the image, whereas the crater rim is much flatter and the rim crest becomes too difficult to identify (see Text S4 and Figures S5–S6 in Supporting Information S1) (Turtle et al., 2005; Xie et al., 2017). As a result, the initial crater diameter is only $\sim 4\%$ larger than the crater diameter measured in the optical image. Given that the manual determination of crater diameter in optical images has an uncertainty of about 10% (Robbins et al., 2014), the above underestimation and overestimation of crater diameters are canceled out. Therefore, no further correction is needed for the crater diameter measured in the TC image of unit Em4, and the crater excavation depth can be directly estimated as 0.084 times the crater diameter measured in imagery data (see Equation 1).

3.2. Morphology of Partially Buried Craters

In this study, we mapped 37 craters whose distal ejecta were buried by lava flows (Figure 8 and Table 2). Their spatial distributions show that the majority of them are located along mare-highland boundaries in the north and east, where basalts are thinner and impact craters are more difficult to be completely buried. Among all 37 buried craters, three craters have completely exposed rims with diameters of 1.94, 3.19, and 3.73 km (Figure 8b). Twenty-six craters have partially exposed rims with a diameter range of 0.87–9.72 km and a median of 3.87 km (Figure 8c). Eight craters have completely buried rims with diameters of 1.15–2.38 km and a median of 1.54 km (Figure 8d). We then simulated the topographic degradation of three rim completely exposed craters (Figure 9), and our results showed that their elevation profiles were well fitted (RMS misfit < 10 m). In Figure 9a, we demonstrated the topographic evolution history of a typical partially buried crater (40.38°N, 51.22°W, $D = 3.19$ km; Figure 4a). This selected crater initially formed with a diameter of 3.186 km and a rim height of 118 m. After a period of time, the crater diameter increased to 3.2 km and the rim height was eroded to 104 m. Next, the lava flow flooded the crater ejecta with a thickness of 63 m. The crater degradation then continued to the present day, finally leaving a crater diameter of 3.204 km and a rim height of 102 m. Note that the final crater diameter is slightly larger than the observed crater diameter (3.204 vs. 3.19 km), but we have set the maximum initial crater diameter to be smaller than the observed crater diameter in the inversion model (see Text S2 in Supporting Information S1).

As shown in Figure 9d, the TiO_2 content of the crater rim is 2.2–4.5 wt.%, indicating that this crater must have formed on Em2 and was later flooded by Em3/Em4. If the Em2 unit does exist beneath the Em3/Em4 unit, this also indicates that the proximal ejecta (with a TiO_2 content of 2.2–4.5 wt.%) of Class A penetrating craters are made of Em2 materials, instead of being formed by mixing Em3/Em4 and Im2 that have higher (> 4.5 wt.%) and lower (1.2–2.2 wt.%) TiO_2 contents. The basalt thickness (63 m) of Em3/Em4 given by the partially buried crater in Figure 9a is within the thickness range (9–175 m) of Em3/Em4 returned by the Class A penetrating

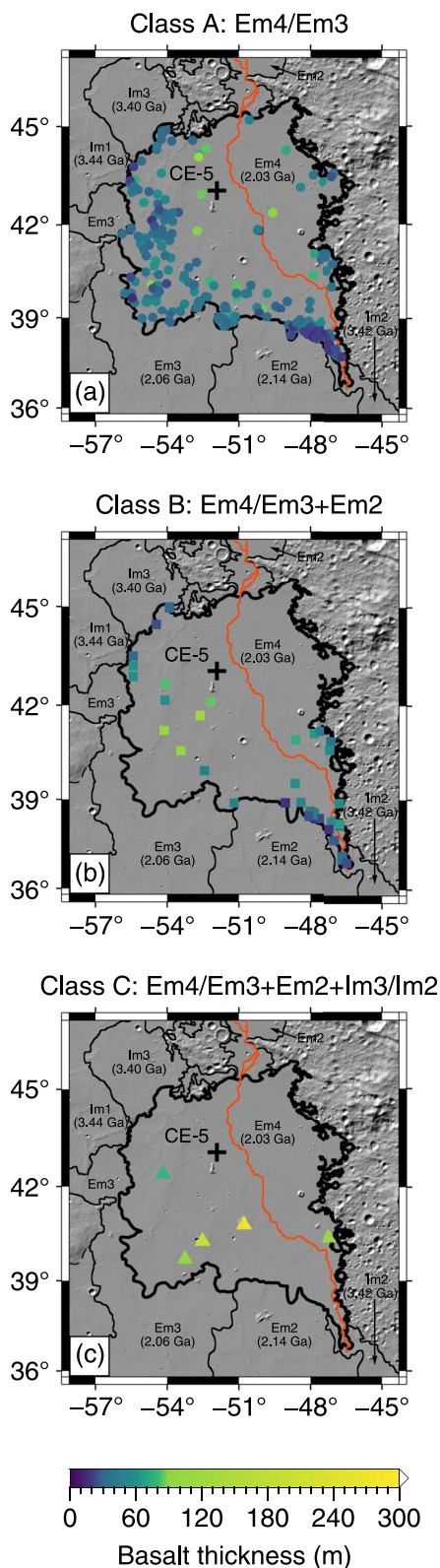


Figure 7. Spatial distributions of summed basalt thicknesses of (a) Em4/Em3 (dots), (b) Em4/Em3+Em2 (squares), and (c) Em4/Em3+Em2+Im3/Im2 (triangles) units estimated by excavation depths of Class A, B, and C penetrating and nonpenetrating craters, respectively. The base map is the LRO/Lunar Orbiter Laser Altimeter (LOLA) shaded relief.

Table 2
Classification of Identified Partially and Completely Buried Craters

Rim morphology	Completely exposed	Partially exposed	Completely buried
Number	3	26	8
Diameter range (km)	1.94–3.73	0.87–9.72	1.15–2.38
Median diameter (km)	3.19	3.87	1.54

and nonpenetrating craters. As a comparison, the TiO_2 content of the crater rim in Figure 9c is 1.2–2.2 wt.%, indicating that this crater must form on Im2/Im3 and was later flooded by Em2 and Em3/Em4. The inverted basalt thickness (43 m) of Em2+Em3/Em4 returned by the partially buried crater in Figure 9b is also within the thickness range (11–167 m) of Em2+Em3/Em4 given by the Class B penetrating and nonpenetrating craters. The third crater is located on the mare-highland boundary (Figure 9f) and the surrounding basalt thickness was estimated to be 51 m (Figure 9c). Our results are only 20%–50% of those obtained from Jia et al. (2021), as their study used much larger partially buried craters (4.5 and 6.5 km) and they did not consider the effect of crater topographic degradation (Du et al., 2019). For those craters

who have a partially exposed or completely buried rim, lower bounds on the basalt thickness of 32–360 m were obtained from the rim height-crater diameter relation of fresh lunar craters without considering crater erosion (Figure 10) (Du et al., 2019). Note that in Figure 10 the basalt thicknesses along the eastern mare-highland boundary are larger than many estimates within the Em4 unit. This is because the former represents the total basalt thickness whereas the latter only refers to the thickness of the upper fraction of total basalts that embayed the crater (Du et al., 2019).

3.3. Effective Density Spectrum

When using the effective density spectrum technique to estimate the basalt thickness, it is first required that the observed free-air gravity be closely correlated with the gravity predicted by topography (Gong et al., 2016). A threshold on their correlation coefficient ($R_0 > 0.775$, corresponding to a signal-to-noise ratio of 1.5) was chosen as the lower limit that later allowed us to fit the effective density spectrum between degrees of 250 and 630 (Figure 11a). A priori constraints on the bulk density of mare basalts ($3,288 \text{ kg/m}^3$) and the density gradient of the highland substrate ($21.2 \text{ kg/m}^3/\text{km}$), which were obtained from the ilmenite abundance of the landing region (Huang & Wiczorek, 2012) and gravity observations of lunar highlands (Besserer et al., 2014), were then utilized to reduce the number of free parameters in our inversion. As a result, we were able to look for the smallest misfit in a grid search manner by only varying the total basalt thickness and the density of the first layer in the highland substrate. When performing the multitaper spectrum analyses, a total number of 30 tapers were selected (Figure S11 in Supporting Information S1) with (a) the geometric center located at the Chang'E-5 landing site, (b) an angular radius of 8° that covers the Em4 unit, and (c) a spherical harmonic bandwidth of 106. In the end, the modeled effective density spectrum (green) that best fits the observed effective density spectrum (red) yielded a total basalt thickness of 340 m (Figure 11b), and the maximum allowable misfit confined a thickness uncertainty of 0–800 m (Figure 11c).

It should be noted that any interpretation of the total basalt thickness inverted from the gravity data needs additional caution. This is because this technique assumes a basalt layer with uniform thickness and density, and as a result, the resulting basalt thickness estimate actually represents an average total basalt thickness of a large region sampled by the tapers. In addition, the tapers chosen here are about five times the size of the Em4 unit and they also cover the volcanic center Mons Rümker and the Imbrium ejecta Montes Jura (Figure 5), which may lead to significantly large and small basalt thickness estimates, respectively. However, if we chose a taper with an angular radius smaller than 8° , we ended up with fewer ($K < 30$) qualified (with 99% of the spectral power concentrated within the angular radius) tapers that further lead to large variations in the observed effective density spectrum. Moreover, we made use of a simplified density profile for the highland substrate but the actual highland substrate could be more heterogeneous in its density structure (Richardson & Abramov, 2020). Taken together, the effective density spectrum model thus returned a total basalt thickness with a large variation of 0–800 m.

As can be seen in the Kaguya/MI FeO content map of the Chang'E-5 landing region (Figure S12 in Supporting Information S1), none of the craters within the study area have excavated materials from the underlying highland substrate (with a FeO content of $<11.5 \text{ wt.}\%$, similar to the nearby highland region Montes Jura). Therefore, the excavation depth (470 m) of the largest crater (Mairan G: 40.90°S , 50.83°W ; $D = 5.69 \text{ km}$) can serve as a lower bound of the total basalt thickness, which is comparable to the best fitting total basalt thickness of 340 m derived from gravity observations. In addition, this total basalt thickness obtained from gravity inversions is significantly larger than that estimated by the partially buried crater at the mare margin (51 m), which is to be expected as the

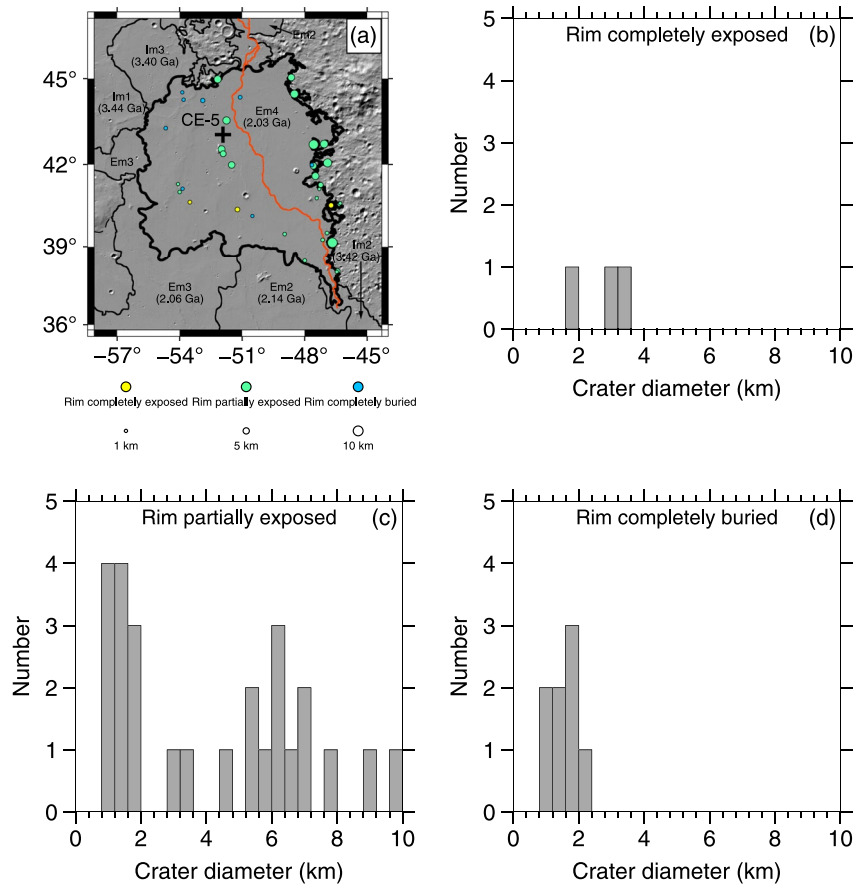


Figure 8. (a) Spatial distributions of buried craters with completely exposed rims (yellow), partially exposed rims (green), and completely buried rims (blue). The crater diameters are scaled for visibility. Diameter histograms of buried craters with (b) completely exposed rims, (c) partially exposed rims, and (d) completely buried rims.

lava flow along the mare-highland boundary is much thinner (Du et al., 2019). In this study, we used the thickness estimate obtained from gravity inversions as the total basalt thickness of the Chang'E-5 landing region as it is more representative for the entire study area.

3.4. Summary of Basalt Thickness Estimates

At this point, the basalt thicknesses of the Em4/Em3, Em2, and Im3/Im2 units have been estimated from penetrating/nonpenetrating craters and partially buried craters, and the total basalt thickness was also obtained from gravity inversions. The thickness of the Im1 unit thus can be estimated as the total basalt thickness minus the combined basalt thickness of Em4/Em3+Em2+Im3/Im2 units. For simplicity and as a secondary consideration, the median value, instead of the upper and lower bounds, of the basalt thicknesses of each unit is used from here for geological interpretations. To account for the variation of the basalt thickness in the region, the standard deviation of the basalt thickness is also used below. By doing so, we can separate the thickness of Em2 from the combined thickness of Em4/Em3+Em2 and can subtract the thickness of Im3/Im2 from the summed thickness of Em4/Em3+Em2+Im3/Im2. This further enables us to present the local stratigraphic column of the Chang'E-5 landing region in layers (Figure 12) and calculate the eruption rate of mare basalts (Section 4.1). As a result, the basalt thicknesses of Em4/Em3, Em2, Im3/Im2, and Im1 units were estimated to be 36^{+17}_{-17} m, 4^{+16}_{-4} m, 70^{+76}_{-70} m, and 230^{+392}_{-230} m, respectively. Based on the obtained thicknesses and surface ages of the basaltic units, we are now able to depict the history of volcanic activities in the Chang'E-5 landing region as follows (Figure 12). In the northeastern Oceanus Procellarum, the earliest and the most violent volcanic activities took place during the Imbrian epoch from 3.44 to 3.40 Ga (Hiesinger et al., 2011), and the resulting lava flows formed the Im1 and Im2/Im3 units with thicknesses of 230^{+392}_{-230} m and 70^{+76}_{-70} m, respectively. After this, the volcanic eruption in this region turned into a much

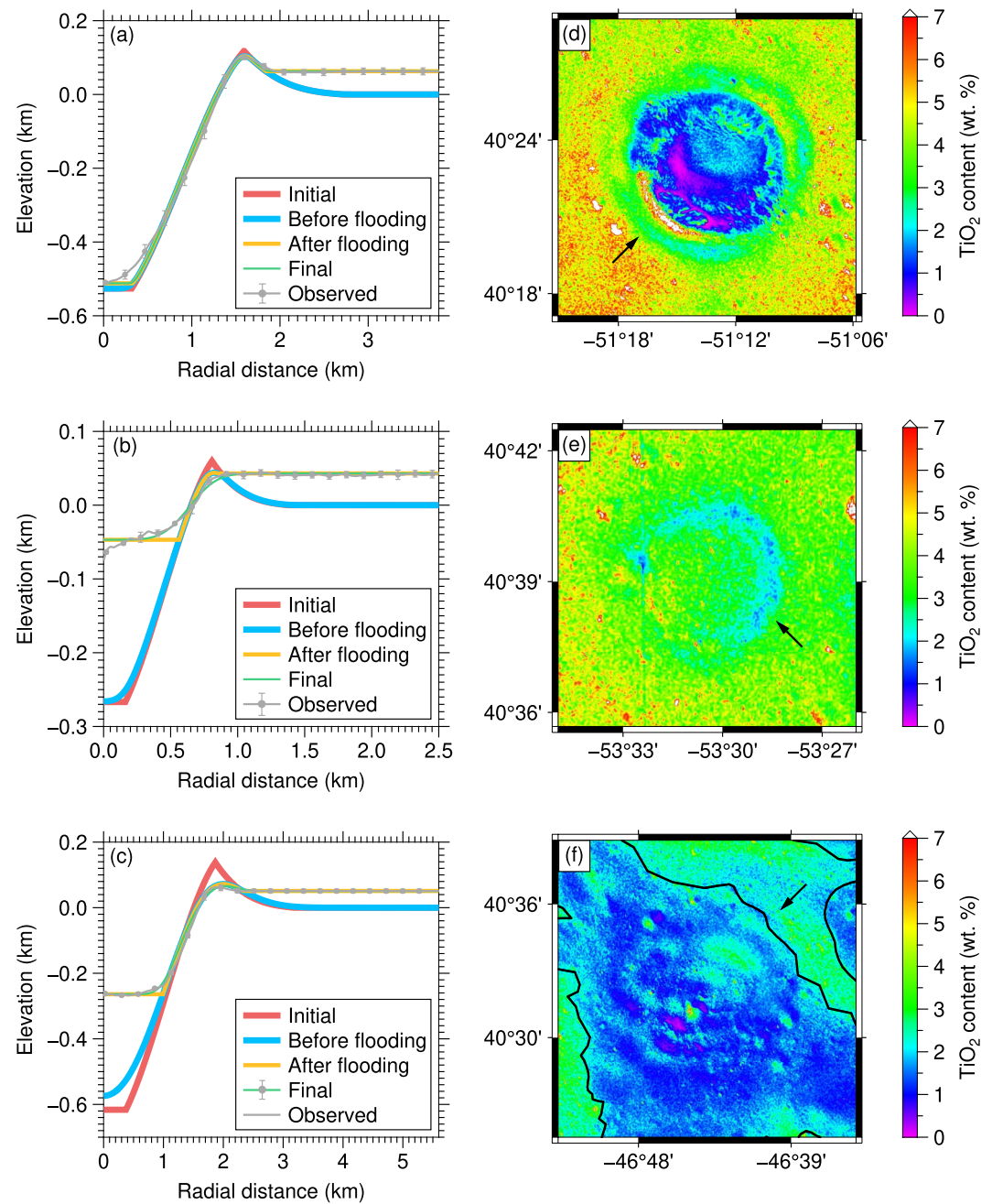


Figure 9. (a–c) Topographic evolution and (d–f) TiO₂ contents of all three partially buried craters with completely exposed rims (see Figures 4a–4c). In (a–c), the red, blue, orange, green, and gray curves represent the initial, before flooding, after flooding, final, and observed elevation profiles. In (f), the mare-highland boundary is shown by black curves. Note that the color bars in (d–f) are the same as the one used in Figure 1c.

weaker mode at the beginning of the Eratosthenian epoch, during which mare basalts with a thickness of only 4^{+16}_{-4} m erupted by 2.14 Ga (Hiesinger et al., 2011). Mare volcanism in this area was then reactivated, giving rise to the emplacement of the Em3 and Em4 units with a summed thickness of 36^{+17}_{-17} m until 2.03 Ga (Q. Li et al., 2021).

The spatial distributions of basalt thickness estimates obtained from penetrating/nonpenetrating craters (Figure 7) and partially buried craters (Figure 10) demonstrate a clear thickening trend from the unit boundary to Rima Sharp and Rima Mairan. This finding is also consistent with the GRAIL Bouguer gravity anomaly map of this region (Figure 5c), which shows positive gravity signals around the two sinuous rilles most likely due to the presence of

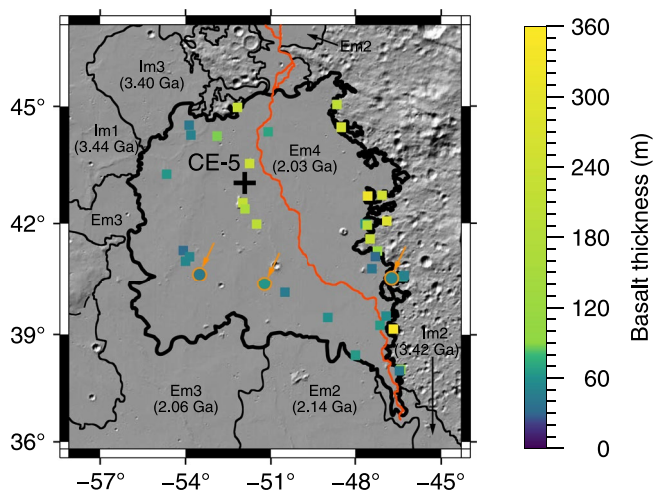


Figure 10. Spatial distribution of basalt thicknesses obtained from partially and completely buried craters. Estimates obtained from three partially buried craters with completely exposed rims are denoted by orange circles and arrows. Lower bounds of basalt thicknesses obtained from craters with buried rims are shown in squares.

dense mare basalts (Evans et al., 2016; Neumann et al., 2015). For the Erato-sthenian basalts within the Em4 unit, it can be inferred that they were fed by Rima Sharp and Rima Mairan whose source vents are located to the north and south of the Em4 unit (Qian, Xiao, Head, & Wilson, 2021). The paths of the two sinuous rilles were likely guided by preexisting surface slopes into regions with lower elevations and then merged with each other. For the terrains with a low topography, they might have formed due to the loading of much thicker Imbrian lava flows and the subsequent crust subsidence. There is little evidence on the surface, however, for the origin of the Imbrian basalts under the Em4 unit. Nevertheless, previous geological mapping of the Imbrium Basin showed that the fourth ring of the Imbrium Basin is close to the region with positive Bouguer gravity anomalies (Qian, Xiao, Head, & Wilson, 2021; Spudis et al., 1988) (Figure 5c). The multiring system of lunar basins was interpreted to form as a result of normal faults that arose from the collapse of the transient crater (Johnson et al., 2016; Melosh, 1989). Therefore, it is likely that the faulted zone beneath the Imbrium rings provided favorable stress conditions for the magma ascent during the Imbrian Epoch, which later resulted in the localization of mare basalts in the region with positive Bouguer gravity anomalies (Michaut & Pinel, 2018).

4. Discussions

4.1. Eruption Rates of Mare Basalts and Thermal Evolution of the Moon

We next calculated the eruption rate of mare basalts in the Chang'E-5 landing region, which could allow us to better quantify and understand the late-stage volcanism on the Moon. For a given basaltic unit, we made use of the median

value, instead of upper and lower limits, of the thickness estimates as a first-order approximation to the basalt thickness of this unit. In addition, as the basalt thickness is not uniform across the entire unit, we also used the standard deviation of the thickness estimates to represent the variation of the basalt thickness in the study region. These two steps greatly simplified the procedures to calculate the eruption rate but can still help us to derive robust geological implications about the thermal evolution history of the Moon. The eruption rate can be defined by either the thickness or the volume of mare basalts, and we calculated the eruption rate using both definitions, although the latter was more commonly reported in previous studies (Du et al., 2019; Oshigami et al., 2014; Weider et al., 2010). This is because (a) the volume is a measure of total amount of mare basalts emplaced on the lunar surface, which, in the lunar thermochemical modeling, can serve as a critical constraint on the total amount of partial melts produced in the mantle, and (b) the thickness of mare basalts is highly dependent on the preexisting topography with deep structures (e.g., basin floors) prone to accommodating thicker deposits.

For each basaltic unit, the thickness was derived from this study, the surface age was taken from Hiesinger et al. (2011) or Q. Li et al. (2021), and the total area was assumed to be its exposed surface area added by the surface areas of all its overlying units. In previous attempts to calculate the total volume of a given basaltic unit (e.g., Oshigami et al., 2014; Weider et al., 2010), we noted that the portion of the basalts that were buried by subsequent lava flows was always neglected, which results in an underestimation of the total basalt volume. Then, assuming that the unit A has a volume of V and a thickness of T and was erupted at time t_i and that the unit B has an age of t_{i-1} and underlies the unit A (see Figure 12), the eruption rate was calculated as $V/(t_i - t_{i-1})$ and $T/(t_i - t_{i-1})$. The age of the highland substrate that underlies Im1 was assumed to be the age of the lunar highland crust (4.55 Ga) (Shearer et al., 2006). By this definition, it therefore should be noted that the resulting eruption rate represents the long-term average flux of magmatic eruptions, which reveals the lunar thermal history over geologic time. To calculate the instantaneous eruption rate of mare basalts, we need more geochemical and rheological evidence from the basalt samples, which is beyond the scope of this study. Combined with previous estimates (Oshigami et al., 2014; Weider et al., 2010), it can be seen in Figure 13 that the eruption rate (no matter defined by thickness or volume) first reached a peak at 3.4 Ga and then flattened from 3.2 to 2.7 Ga, which represents the major phase of volcanic eruptions on the Moon. After this, there was a significant drop in the eruption rate between 2.7 and 2.1 Ga and there are few estimates during this period of time, indicating that lunar volcanic

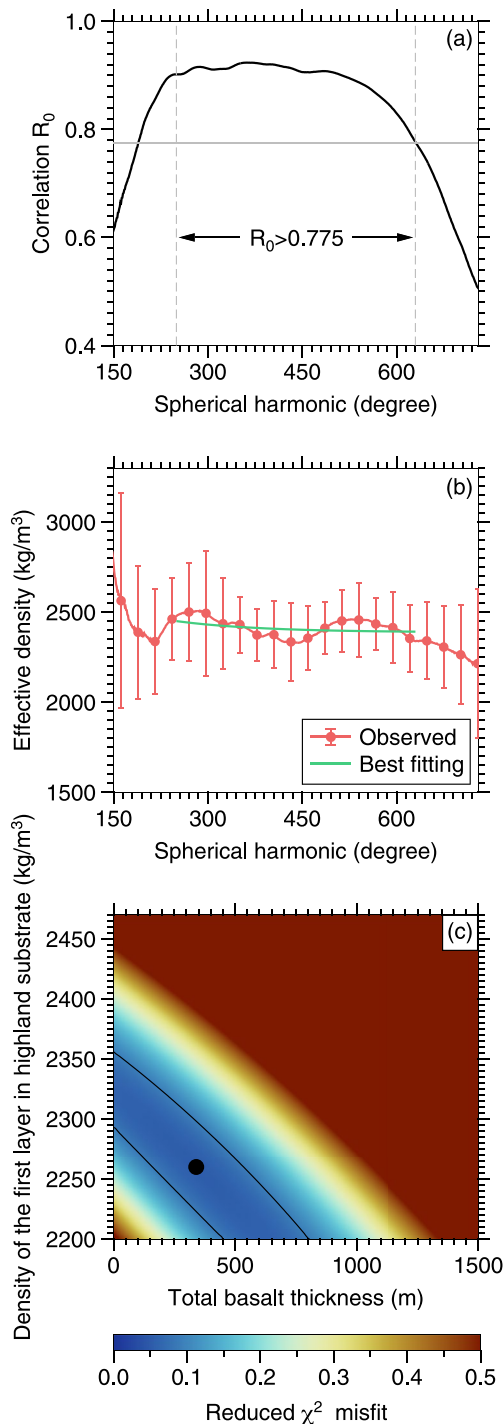


Figure 11. (a) Correlation coefficient between the observed free-air gravity and the gravity predicted by topography. A threshold of 0.775 between degrees 250 and 630 is shown. (b) Observed (red) and the best fitting modeled (green) effective density spectra. (c) Reduced χ^2 misfit of the effective density spectrum as a function of the total basalt thickness and the density of the first layer in the highland substrate. The best fitting total basalt thickness is given by the black dot with its uncertainty denoted by the black curves.

activities switched to a dormant mode. At 2.0 Ga, the eruption rate rose again, reaching to the same level as shown during 3.2–2.7 Ga (Figure 13). It should be noted that this uptick in the eruption rate of mare basalts at around 2.0 Ga has also been reported in other works. Du et al. (2019) estimated the basalt thicknesses on lunar maria based on the morphology of partially buried craters and found an increase in the calculated magmatic flux at 1.7 Ga. Both Hiesinger et al. (2011) and Morota et al. (2011) dated the basaltic units on the lunar nearside, and the resulting histograms of their model ages also revealed a secondary peak at 1.8–2.2 Ga.

Although six ages were cited in this study, only four of them (Im1, Im3 and Em2, Em4) were used here to calculate the eruption rates of mare basalts. As mentioned above, the eruption rate was defined as the amount of mare basalts that reached to the lunar surface divided by the time length $t_i - t_{i-1}$, where t_i and t_{i-1} are the ends of the previous eruption and this eruption, respectively. Therefore, after we merged Im2 and Im3 to Im2/Im3 and combined Em3 and Em4 to Em3/Em4, we actually only used the age of Im3 to represent that of Im2/Im3 and the age of Em4 to represent that of Em3/Em4. As a result, among the four ages used to calculate the eruption rates, three of them (Im1, Im3, and Em2) were crater-counting ages from Hiesinger et al. (2011) and hence should be self-consistent, whereas the radiometric age of Em4 from Q. Li et al. (2021) may have a systematic bias compared with the other three: The radiometric age was obtained from returned samples but the crater-counting age represents the age of a much larger region. However, we found that even if we used the crater-counting age of Em4 (1.33 Ga) from Hiesinger et al. (2011), the last eruption rate would drop from 2.9×10^4 to 3.9×10^3 km³/Gyr, which is still 10 times higher than the previous eruption rate (405 km³/Gyr) at 2.14 Ga. Therefore, the systematic bias between ages obtained from crater counting and radiometric dating does not change our conclusion that there is an uptick in the recent lunar volcanic activities.

Previous studies ascribed the rejuvenated late-stage volcanism on the Moon at around 2 Ga to the enrichment of heat-producing elements (e.g., U, Th, and K) in the hypothesized KREEP layer under the Procellarum KREEP Terrane (e.g., Laneuville et al., 2013; Wieczorek & Phillips, 2000). In a recent sophisticated thermochemical modeling of the Moon (Laneuville et al., 2018), the PKT region was divided into an inner portion with an enriched thorium content of 8.2 ppm and an outer portion with a lower thorium concentration of 4.2 ppm, where the Em4 unit straddles these two subregions. This initial distribution of heat-producing elements in the PKT was set up based on the acquired thorium content (Gillis et al., 1999) and the magnetic field strength (Wieczorek, 2018) in this region, which is also in accordance with the calculated thorium content of the Chang'E-5 basalts (4.5 ppm) (Tian, Wang, Chen, et al., 2021). Their simulation then showed that the melt production in the lunar mantle could extend to 2 Ga, which is consistent with the time span of the nearside volcanism on the Moon given by the radiometric dating of the Chang'E-5 basalts (Q. Li et al., 2021). In addition, Laneuville et al. (2018) also succeeded to reproduce episodic enhancements of melt production at 3.5 and 2.3 Ga that might result from the upwelling hot mantle plumes, and the latter uptick at 2.3 Ga may correspond to the Erastothernian volcanic eruptions in the PKT region. We noted that, however, a direct comparison of absolute melt volumes between their results and our estimates is quite challenging. This is because the total volume of mare basalts derived from the thermochemical model strongly depends on the solidus of lunar mantle

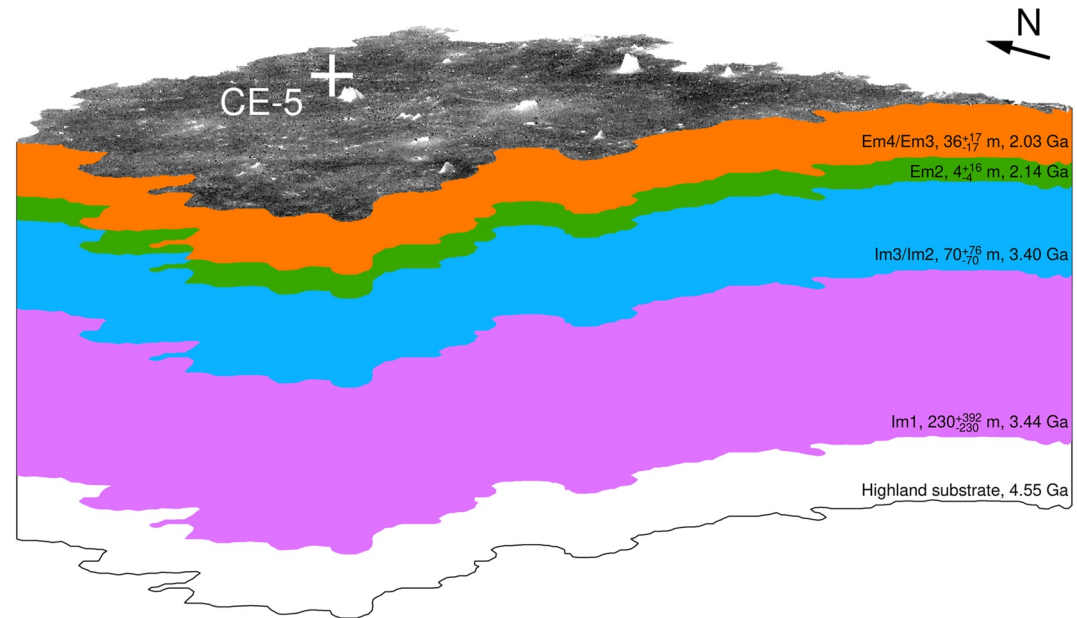


Figure 12. Stratigraphic column of the Chang'E-5 landing region. From the lunar surface down to the lunar highland substrate, the Em4/Em3 (orange), Em2 (green), Im3/Im2 (blue), and Im1 (purple) units are shown with their thicknesses (not to scale) and ages denoted. The Chang'E-5 landing site (white cross) is marked on the surface where the Kaguya/TC image of the Em4 unit is rendered by the Kaguya/TC DTM.

and the intrusion-extrusion ratio of magmas on the Moon that are both poorly constrained a priori (Laneuville et al., 2018).

4.2. Implications From Latest Chang'E-5 Sample Analyses

In contrast to the thermochemical modeling efforts (e.g., Laneuville et al., 2013, 2018; Wieczorek & Phillips, 2000), the contribution of the KREEP layer to the prolonged mare volcanism was discredited by the latest isotopic measurements of the returned Chang'E-5 basalt samples. The calculated $^{238}\text{U}/^{204}\text{Pb}$ (677–684), $^{87}\text{Rb}/^{86}\text{Sr}$ (0.009–0.022), and $^{147}\text{Sm}/^{144}\text{Nd}$ (0.222–0.227) ratios of the Chang'E-5 basalts clearly deviate from what is expected for KREEP basalts (>1,000, 0.195, and 0.173) (Che et al., 2021; Q. Li et al., 2021; Tian, Wang, Chen, et al., 2021). In addition to this, it was also reported that the Chang'E-5 basalts have a Th content of 4.5 ppm (Tian, Wang, Chen, et al., 2021), which is significantly lower than the Apollo 15 KREEP basalt with an average Th concentration of 12.3 ppm (Korotev, 2000). Taken together, these results indicated that neither the mantle source of the Chang'E-5 basalts is enriched in KREEP components nor the Chang'E-5 basalts were mixed with KREEP materials during their ascent through the crust, which downplayed the prerequisite of a KREEP layer to sustain the long-term volcanism on the Moon (Che et al., 2021; Q. Li et al., 2021; Tian, Wang, Chen, et al., 2021). Instead, Che et al. (2021) proposed that tidal forces exerted by the Earth could provide extra heat for the lunar interior. Tian, Wang, Chen, et al. (2021) pointed out that a thinner crust layer in PKT (<30 km vs. a global average of 34–43 km) (Wieczorek et al., 2013), where uprising magmas experienced shorter paths of negative buoyancy, may also contribute to the long lasting nearside volcanism. The elevated abundance of thorium (5–8.5 ppm) observed in the Em4 unit could be explained by the presence of highly evolved magmas, which experienced a low-degree (2%–3%) partial melting followed by an extensive (43%–78%) fractional crystallization (Tian, Wang, Chen, et al., 2021), or alternatively, as a result of the lateral mixing of high-Th Imbrium ejecta (Zhang et al., 2021).

In our opinion, it can be expected to have obtained an extremely low abundance of KREEP materials for the mantle source of the Em4 basalts. This is because (a) the lunar mantle was believed to be depleted in the incompatible elements after the crystallization of the lunar magma ocean (Warren, 1985) and (b) the thermochemical model shows that the KREEP layer is stratigraphically stable and thus is not involved in the underlying mantle convection (Laneuville et al., 2018). By assuming a moderately enriched crust and a highly depleted mantle with respective thorium contents of 4.2 and 0.055 ppm, the result from thermochemical modeling actually proved

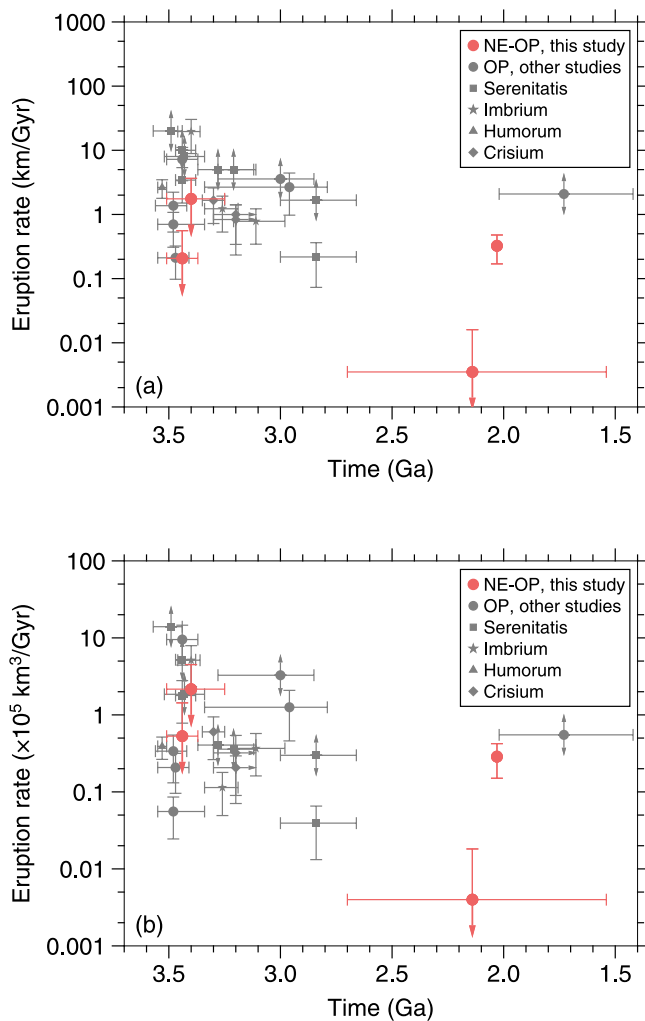


Figure 13. Eruption rates of mare basalts defined by (a) thickness and (b) volume in the Chang'E-5 landing region (red) in comparison to those in other mare plains (gray) (Oshigami et al., 2014; Weider et al., 2010). NE-OP stands for the northeastern Oceanus Procellarum. Note that the surface areas used in Weider et al. (2010) and Oshigami et al. (2014) have been updated. Originally, Weider et al. (2010) mistakenly used the much smaller crater-counting areas, and Oshigami et al. (2014) may not use an equal-area projection system to calculate the surface area.

19–478, and 0–800 m, respectively. Combined with the latest radiometric and crater-counting ages of the basaltic units, we calculated the eruption rate of mare basalts in the Chang'E-5 landing region. Our results revealed an uptick in the magmatic flux at about 2 Ga, suggesting the presence of rejuvenated volcanic activities on the Moon. The long-lived volcanism in the northeastern Oceanus Procellarum can be explained as a result of the enriched heat-producing elements in the crust, or alternatively, due to tidal heating and/or a relatively thin crust layer.

Data Availability Statement

The Kaguya/MI and TC data can be found at the JAXA Data Archive (https://darts.isas.jaxa.jp/planet/pdap/selene/product_search.html). To download the Kaguya/MI images, select the Data set ID as SLN-L-MI-5-MAP-V3.0 and then enter the range of the study region (35°N–49°N, 42°W–61°W). To download the Kaguya/TC images,

that lunar volcanism can sustain until ~2 Ga (Laneuville et al., 2018). The major concern raised by the isotopic analyses of the Chang'E-5 basalts is that the returned basaltic samples were not mixed with the KREEP layer during their ascent to the lunar surface (Che et al., 2021; Q. Li et al., 2021; Tian, Wang, Chen, et al., 2021). It is also likely, however, that the magmas ascended quickly through the dikes without assimilating with the surrounding KREEP-rich rocks. A feasible way to examine if this assimilation process has occurred would be looking at the zoning pattern of the olivine crystal in the Chang'E-5 basaltic clasts through the electron microprobe traverse technique. If the Chang'E-5 basalts were not assimilated with the KREEP materials, we are expected to see a typical normal zoning pattern within the crystal.

From the perspective of thermochemical modeling, it is suggested that future analyses of Chang'E-5 samples could be focused on their geophysical properties. The relevant model parameters include solidus, density, viscosity, heat capacity, and thermal conductivity of mantle materials (Laneuville et al., 2013, 2018). With better constraints on these model parameters and reliable estimates of basalt thicknesses, the thermochemical model can be improved and used to investigate the driving force of the young volcanism on the Moon. In addition to this, one issue that remains unsolved in our study is the thickness of the Em4 unit. Potentially, this could be answered by looking into the cooling history of the returned Chang'E-5 basalts. As mare basalts cooled with time, the growth of crystals (e.g., olivine and pyroxene) leads to a unique pattern of compositional zoning. By measuring the concentration traverse profiles of certain compositions (e.g., Fe, Mn, and Ni), the cooling rate of mare basalts can then be calculated (Day & Taylor, 2007; Richter et al., 2021). Using the resulting cooling rate as a constraint, a thermal diffusion model of the lava flow can thus be used to invert the thickness of the Em4 unit, which allows us to refine the stratigraphic column obtained in this study.

5. Conclusions

In this study, we estimated the basalt thicknesses in the Chang'E-5 landing region based on the crater excavation depths of penetrating/nonpenetrating craters, the morphology of partially buried craters, and the effective density spectrum of gravity and topography. In comparison to previous studies, we (a) made a more detailed classification of penetrating craters, (b) considered the topographic degradation of partially buried craters, and (c) utilized the most up-to-date gravity model for inversions. Our results showed that the Em4/Em3, Em2, Im3/Im2, and Im1 basaltic units were emplaced in the Chang'E-5 landing region with accumulated thicknesses of 9–175, 11–167,

select the Data set ID as SLN-L-TC-5-DTM-MAP-SEAMLESS-V2.0 and then enter the range of the study region. The GRAIL data are available at the NASA Planetary Geodynamics Data Archive (<https://pgda.gsfc.nasa.gov/products/75>) and in Kahan (2013). The LRO/LOLA spherical harmonics coefficients can be found in Wieczorek (2007). Basalt thickness estimates obtained in this study, database of penetrating and nonpenetrating craters and inversion results of their topographic degradation modeling, and database of partially and completely buried craters and inversion results of their topographic degradation modeling are accessible at Du et al. (2022).

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References

- Andrews-Hanna, J. C., Asmar, S. W., Head, J. W., III, Kiefer, W. S., Konopliv, A. S., Lemoine, F. G., et al. (2013). Ancient igneous intrusions and early expansion of the Moon revealed by GRAIL gravity gradiometry. *Science*, 339(6120), 675–678. <https://doi.org/10.1126/science.1231753>
- Besserer, J., Nimmo, F., Wieczorek, M. A., Weber, R. C., Kiefer, W. S., McGovern, P. J., et al. (2014). GRAIL gravity constraints on the vertical and lateral density structure of the lunar crust. *Geophysical Research Letters*, 41(16), 5771–5777. <https://doi.org/10.1002/2014GL060240>
- Che, X., Nemchin, A., Liu, D., Long, T., Wang, C., Norman, M. D., et al. (2021). Age and composition of young basalts on the Moon, measured from samples returned by Chang'E-5. *Science*, 374(6569), 887–890. <https://doi.org/10.1126/science.abl7957>
- Day, J. M., & Taylor, L. A. (2007). On the structure of mare basalt lava flows from textural analysis of the LaPaz Icefield and Northwest Africa 032 lunar meteorites. *Meteoritics & Planetary Sciences*, 42(1), 3–17. <https://doi.org/10.1111/j.1945-5100.2007.tb00213.x>
- Du, J., Fa, W., Gong, S., Liu, Y., Qiao, L., Tai, Y., & Zou, Y. (2022). Thicknesses of mare basalts in the Chang'E-5 landing region: Implications for the late-stage volcanism on the Moon. *Zenodo*. <https://doi.org/10.5281/zenodo.6960322>
- Du, J., Fa, W., Wieczorek, M. A., Xie, M., Cai, Y., & Zhu, M.-H. (2019). Thickness of lunar mare basalts: New results based on modeling the degradation of partially buried craters. *Journal of Geophysical Research: Planets*, 124(9), 2430–2459. <https://doi.org/10.1029/2018JE005872>
- Evans, A. J., Soderblom, J. M., Andrews-Hanna, J. C., Solomon, S. C., & Zuber, M. T. (2016). Identification of buried lunar impact craters from GRAIL data and implications for the nearside maria. *Geophysical Research Letters*, 43(6), 2445–2455. <https://doi.org/10.1002/2015GL067394>
- Fu, X., Hou, X., Zhang, J., Li, B., Ling, Z., Jolliff, B. L., & Zou, Y. (2021). Possible non-mare lithologies in the regolith at the Chang'E-5 landing site: Evidence from remote sensing data. *Journal of Geophysical Research: Planets*, 126(5), e2020JE006797. <https://doi.org/10.1029/2020JE006797>
- Gillis, J. J., Haskin, L. A., & Spudis, P. D. (1999). An empirical calibration to calculate thorium abundances from the lunar prospector gamma-ray data. In *The 30th lunar and planetary science conference*. Abstract 1699.
- Gong, S., Wieczorek, M. A., Nimmo, F., Kiefer, W. S., Head, J. W., Huang, C., et al. (2016). Thicknesses of mare basalts on the Moon from gravity and topography. *Journal of Geophysical Research: Planets*, 121(5), 854–870. <https://doi.org/10.1002/2016JE005008>
- Goossens, S., Sabaka, T. J., Wieczorek, M. A., Neumann, G. A., Mazarico, E., Lemoine, F. G., et al. (2020). High-resolution gravity field models from GRAIL data and implications for models of the density structure of the Moon's crust. *Journal of Geophysical Research: Planets*, 125(2), e2019JE006086. <https://doi.org/10.1029/2019JE006086>
- Haruyama, J., Hara, S., Hioki, K., Iwasaki, A., Morota, T., Ohtake, M., et al. (2012). Lunar global digital terrain model dataset produced from SELENE (Kaguya) Terrain Camera stereo observations. In *The 43rd lunar and planetary science conference*. Abstract 1200.
- Haruyama, J., Matsunaga, T., Ohtake, M., Morota, T., Honda, C., Yokota, Y., et al. (2008). Global lunar-surface mapping experiment using the Lunar Imager/Spectrometer on SELENE. *Earth Planets and Space*, 60(4), 243–255. <https://doi.org/10.1186/BF03352788>
- Hiesinger, H., Head, J. W., Wolf, U., Jaumann, R., & Neukum, G. (2011). Ages and stratigraphy of lunar mare basalts: A synthesis. *Geological Society of America Special Paper*, 477, 1–51. <https://doi.org/10.1130/2011.2477/01>
- Huang, Q., & Wieczorek, M. A. (2012). Density and porosity of the lunar crust from gravity and topography. *Journal of Geophysical Research*, 117(E5), E05003. <https://doi.org/10.1029/2012JE004062>
- Jia, B., Fa, W., Xie, M., Tai, Y., & Liu, X. (2021). Regolith properties in the Chang'E-5 landing region of the Moon: Results from multi-source remote sensing observations. *Journal of Geophysical Research: Planets*, 126(7), e2021JE006934. <https://doi.org/10.1029/2021JE006934>
- Jia, M., Yue, Z., Di, K., Liu, B., Liu, J., & Michael, G. (2020). A catalogue of impact craters larger than 200 m and surface age analysis in the Chang'E-5 landing area. *Earth and Planetary Science Letters*, 541, 116272. <https://doi.org/10.1016/j.epsl.2020.116272>
- Johnson, B. C., Blair, D. M., Collins, G. S., Melosh, H. J., Freed, A. M., Taylor, G. J., et al. (2016). Formation of the Orientale lunar multiring basin. *Science*, 354(6311), 441–444. <https://doi.org/10.1126/science.aag0518>
- Kahan, D. S. (2013). GRAIL Moon LGRS derived gravity science data products V1.0, GRAIL-L-LGRS-5-RDR-V1.0 [Dataset]. NASA Planetary Data System. <https://doi.org/10.17189/1519529>
- Kato, S., Morota, T., Yamaguchi, Y., Watanabe, S.-i., Otake, H., & Ohtake, M. (2017). Magma source transition of lunar mare volcanism at 2.3 ga. *Meteoritics & Planetary Sciences*, 52(9), 1899–1915. <https://doi.org/10.1111/maps.12896>
- Kiefer, W. S., Macke, R. J., Britt, D. T., Irving, A. J., & Consolmagno, G. J. (2012). The density and porosity of lunar rocks. *Geophysical Research Letters*, 39(7), L07201. <https://doi.org/10.1029/2012GL051319>
- Korotev, R. L. (2000). The great lunar hot spot and the composition and origin of the Apollo mafic (“LKFM”) impact-melt breccias. *Journal of Geophysical Research*, 105(E2), 4317–4345. <https://doi.org/10.1029/1999JE001063>
- Laneuville, M., Taylor, J., & Wieczorek, M. A. (2018). Distribution of radioactive heat sources and thermal history of the Moon. *Journal of Geophysical Research: Planets*, 123(12), 3144–3166. <https://doi.org/10.1029/2018JE005742>
- Laneuville, M., Wieczorek, M. A., Breuer, D., & Tosi, N. (2013). Asymmetric thermal evolution of the Moon. *Journal of Geophysical Research: Planets*, 118(7), 1435–1452. <https://doi.org/10.1002/jgre.20103>
- Li, C., Hu, H., Yang, M., Pei, Z., Zhou, Q., Ren, X., et al. (2021). Characteristics of the lunar samples returned by Chang'E-5 mission. *National Science Review*, 9(2). <https://doi.org/10.1093/nsr/nwab188>
- Li, Q., Zhou, Q., Liu, Y., Xiao, Z., Lin, Y., Li, J., et al. (2021). Two billion-year-old volcanism on the Moon from Chang'E-5 basalts. *Nature*, 600(7887), 54–58. <https://doi.org/10.1038/s41586-021-04100-2>
- Lucey, P. G., Blewett, D. T., & Hawke, B. R. (1998). Mapping the FeO and TiO₂ content of the lunar surface with multispectral imagery. *Journal of Geophysical Research*, 103(E2), 3679–3699. <https://doi.org/10.1029/97JE03019>
- Melosh, H. J. (1989). *Impact cratering: A geologic process*. Oxford University Press.
- Michaut, C., & Pinel, V. (2018). Magma ascent and eruption triggered by cratering on the Moon. *Geophysical Research Letters*, 45(13), 6408–6416. <https://doi.org/10.1029/2018GL078150>

- Moore, H. J., Hodges, C. A., & Scott, D. H. (1974). Multiringed basins-illustrated by Orientale and associated features. In *The 5th lunar and planetary science conference* (pp. 71–100).
- Morota, T., Haruyama, J., Ohtake, M., Matsunaga, T., Honda, C., Yokota, Y., et al. (2011). Timing and characteristics of the latest mare eruption on the Moon. *Earth and Planetary Science Letters*, 302(3), 255–266. <https://doi.org/10.1016/j.epsl.2010.12.028>
- Nava, R. A. (2011). *Crater Helper tools for ArcGIS 10.0 (U.S. Geological survey reference manual)*. U.S. Geological Survey.
- Neukum, G., Ivanov, B., & Hartmann, W. (2001). Cratering records in the inner Solar System in relation to the lunar reference system. *Space Science Review*, 96(1–4), 55–86. <https://doi.org/10.1023/A:1011989004263>
- Neumann, G. A., Zuber, M. T., Wieczorek, M. A., Head, J. W., Baker, D. M., Solomon, S. C., et al. (2015). Lunar impact basins revealed by gravity Recovery and interior laboratory measurements. *Science Advances*, 1(9), e1500852. <https://doi.org/10.1126/sciadv.1500852>
- Ohtake, M., Haruyama, J., Matsunaga, T., Yokota, Y., Morota, T., Honda, C., et al. (2008). Performance and scientific objectives of the SELENE (Kaguya) multiband imager. *Earth Planets and Space*, 60(4), 257–264. <https://doi.org/10.1186/BF03352789>
- Oshigami, S., Watanabe, S., Yamaguchi, Y., Yamaji, A., Kobayashi, T., Kumamoto, A., et al. (2014). Mare volcanism: Reinterpretation based on Kaguya lunar radar sounder data. *Journal of Geophysical Research: Planets*, 119(5), 1037–1045. <https://doi.org/10.1002/2013JE004568>
- Otake, H., Ohtake, M., & Hirata, N. (2012). Lunar iron and titanium abundance algorithms based on SELENE (Kaguya) Multiband Imager data. In *The 43rd lunar and planetary science conference*. Abstract 1905.
- Qian, Y., Xiao, L., Head, J. W., van der Bogert, C. H., Hiesinger, H., & Wilson, L. (2021). Young lunar mare basalts in the Chang'E-5 sample return region, northern Oceanus Procellarum. *Earth and Planetary Science Letters*, 555, 116702. <https://doi.org/10.1016/j.epsl.2020.116702>
- Qian, Y., Xiao, L., Head, J. W., & Wilson, L. (2021). The long sinuous rille system in northern Oceanus Procellarum and its relation to the Chang'e-5 returned samples. *Geophysical Research Letters*, 48(11), e2021GL092663. <https://doi.org/10.1029/2021GL092663>
- Qian, Y., Xiao, L., Wang, Q., Head, J. W., Yang, R., Kang, Y., et al. (2021). China's Chang'E-5 landing site: Geology, stratigraphy, and provenance of materials. *Earth and Planetary Science Letters*, 561, 116855. <https://doi.org/10.1016/j.epsl.2021.116855>
- Qian, Y., Xiao, L., Zhao, S., Zhao, J., Huang, J., Flahaut, J., et al. (2018). Geology and scientific significance of the Rümker region in northern Oceanus Procellarum: China's Chang'E-5 landing region. *Journal of Geophysical Research: Planets*, 123(6), 1407–1430. <https://doi.org/10.1029/2018JE005595>
- Qiao, L., Chen, J., Xu, L., Wan, S., Cao, H., Li, B., & Ling, Z. (2021). Geology of the Chang'E-5 landing site: Constraints on the sources of samples returned from a young nearside mare. *Icarus*, 364, 114480. <https://doi.org/10.1016/j.icarus.2021.114480>
- Qiao, L., Xu, L., Yang, Y., Xie, M., Chen, J., Fang, K., & Ling, Z. (2021). Cratering records in the Chang'E-5 mare unit: Filling the “age gap” of the lunar crater chronology and preparation for its recalibration. *Geophysical Research Letters*, 48(22), e2021GL095132. <https://doi.org/10.1029/2021GL095132>
- Richardson, J. E., & Abramov, O. (2020). Modeling the formation of the lunar upper megaregolith layer. *The Planetary Science Journal*, 1(2), 1–18. <https://doi.org/10.3847/PSJ/ab7235>
- Richter, F., Saper, L. M., Villeneuve, J., Chaussidon, M., Watson, E. B., Davis, A. M., & Simon, S. B. (2021). Reassessing the thermal history of martian meteorite Shergotty and Apollo mare basalt 15555 using kinetic isotope fractionation of zoned minerals. *Geochimica et Cosmochimica Acta*, 295, 265–285. <https://doi.org/10.1016/j.gca.2020.11.002>
- Robbins, S. J., Antonenko, I., Kirchoff, M. R., Chapman, C. R., Fassett, C. I., Herrick, R. R., et al. (2014). The variability of crater identification among expert and community crater analysts. *Icarus*, 234, 109–131. <https://doi.org/10.1016/j.icarus.2014.02.022>
- Sato, B., Robinson, M. S., Lawrence, S. J., Denevi, B. W., Hapke, B., Jolliff, B. L., & Hiesinger, H. (2017). Lunar mare TiO₂ abundances estimated from UV/Vis reflectance. *Icarus*, 296, 216–238. <https://doi.org/10.1016/j.icarus.2017.06.013>
- Shearer, C. K., Hess, P. C., Wieczorek, M. A., Pritchard, M. E., Parmentier, E. M., Borg, L. E., et al. (2006). Thermal and magmatic evolution of the moon. *Reviews in Mineralogy and Geochemistry*, 60(1), 365–518. <https://doi.org/10.2138/rmg.2006.60.4>
- Spudis, P. D., Hawke, B. R., & Lucey, P. G. (1988). Materials and formation of the imbrium basin. In *The 18th lunar and planetary science conference* (pp. 155–168).
- Tian, H., Wang, H., Chen, Y., Yang, W., Zhou, Q., Zhang, C., et al. (2021). Non-KREEP origin for Chang'E-5 basalts in the Procellarum KREEP Terrane. *Nature*, 600(7887), 59–63. <https://doi.org/10.1038/s41586-021-04119-5>
- Turtle, E., Pierazzo, E., Collins, G., Osinski, G., Melosh, H., Morgan, J., & Reimold, W. (2005). Impact structures: What does crater diameter mean? *Geological Society of America Special Paper*, 384, 1–24. <https://doi.org/10.1130/SPE384>
- Warren, P. H. (1985). The magma ocean concept and lunar evolution. *Annual Review of Earth and Planetary Sciences*, 13(1), 201–240. <https://doi.org/10.1146/annurev.ea.13.050185.001221>
- Weider, S. Z., Crawford, I. A., & Joy, K. H. (2010). Individual lava flow thicknesses in Oceanus Procellarum and Mare Serenitatis determined from Clementine multispectral data. *Icarus*, 209(2), 323–336. <https://doi.org/10.1029/2009GL037600>
- Wessel, P., & Smith, W. H. F. (1991). Free software helps map and display data. *EOS Transactions American Geophysical Union*, 72(41), 441–446. <https://doi.org/10.1029/90EO00319>
- Wieczorek, M. A. (2007). Spherical harmonic models of planetary topography. *Zenodo*. <https://doi.org/10.5281/ZENODO.997406>
- Wieczorek, M. A. (2018). Strength, depth, and geometry of magnetic sources in the crust of the Moon from localized power spectrum analysis. *Journal of Geophysical Research: Planets*, 123(1), 291–316. <https://doi.org/10.1002/2017JE005418>
- Wieczorek, M. A., & Meschede, M. (2018). SHTools: Tools for working with spherical harmonics. *Geochemistry, Geophysics, Geosystems*, 19(8), 2574–2592. <https://doi.org/10.1029/2018GC007529>
- Wieczorek, M. A., Neumann, G. A., Nimmo, F., Kiefer, W. S., Taylor, G. J., Melosh, H. J., et al. (2013). The crust of the Moon as seen by GRAIL. *Science*, 339(6120), 671–675. <https://doi.org/10.1126/science.1231530>
- Wieczorek, M. A., & Phillips, R. J. (2000). The “Procellarum KREEP Terrane”: Implications for mare volcanism and lunar evolution. *Journal of Geophysical Research*, 105(E8), 20417–20430. <https://doi.org/10.1029/1999JE001092>
- Wu, B., Huang, J., Li, Y., Wang, Y., & Peng, J. (2018). Rock abundance and crater density in the candidate Chang'E-5 landing region on the Moon. *Journal of Geophysical Research: Planets*, 123(12), 3256–3272. <https://doi.org/10.1029/2018JE005820>
- Xie, M., Zhu, M.-H., Xiao, Z., Wu, Y., & Xu, A. (2017). Effect of topography degradation on crater size-frequency distributions: Implications for populations of small craters and age dating. *Geophysical Research Letters*, 44(20), 10171–10179. <https://doi.org/10.1002/2017GL075298>
- Yang, W., & Lin, Y. (2021). New lunar samples returned by Chang'E-5: Opportunities for new discoveries and international collaboration. *The Innovation*, 2(1), 1–2. <https://doi.org/10.1016/j.xinn.2020.100070>
- Zhang, J. Y., Head, J. W., Liu, J. Z., & Potter, R. W. K. (2021). Analysis of thorium concentration anomalies on the lunar surface. In *The 52nd lunar and planetary science conference*. Abstract 1744.
- Zuber, M. T., Smith, D. E., Watkins, M. M., Asmar, S. W., Konopliv, A. S., Lemoine, F. G., et al. (2013). Gravity field of the moon from the gravity recovery and interior laboratory (GRAIL) mission. *Science*, 339(6120), 668–671. <https://doi.org/10.1126/science.1231507>

References From the Supporting Information

- Culling, W. E. H. (1960). Analytical theory of erosion. *The Journal of Geology*, 68(3), 336–344. <https://doi.org/10.1086/626663>
- Fassett, C. I., & Thomson, B. J. (2014). Crater degradation on the lunar maria: Topographic diffusion and the rate of erosion on the Moon. *Journal of Geophysical Research: Planets*, 119(10), 2255–2271. <https://doi.org/10.1002/2014JE004698>
- MathWorks Inc. (2017). Global optimization toolbox: MATLAB documentation. Retrieved from <https://www.mathworks.com/help/gads/index.html>
- Schittkowski, K. (1986). NLPQL: A FORTRAN subroutine solving constrained nonlinear programming problems. *Annals of Operations Research*, 5(2), 485–500. <https://doi.org/10.1007/BF02022087>
- Wieczorek, M. A., & Phillips, R. J. (1998). Potential anomalies on a sphere: Applications to the thickness of the lunar crust. *Journal of Geophysical Research*, 103(E1), 1715–1724. <https://doi.org/10.1029/97JE03136>
- Xie, M., & Zhu, M.-H. (2016). Estimates of primary ejecta and local material for the Orientale basin: Implications for the formation and ballistic sedimentation of multi-ring basins. *Earth and Planetary Science Letters*, 440, 71–80. <https://doi.org/10.1016/j.epsl.2016.02.012>